PrivAGM: Secure Construction of Differentially Private Directed Attributed Graph Models on Decentralized Social Graphs

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ABSTRACT

Decentralized social graphs, where no single entity possesses the information of the entire graph, and each user maintains only a limited view of the graph, contain great value for different applications. However, simply collecting local views for analytics raises privacy concerns due to the sensitive information of social relationships they capture. To address this, a canonical approach involves privately fitting a generative graph model to the decentralized social graph, generating a differentially private synthetic graph that serves as a proxy for analytics. Existing solutions, however, often fail to capture the inherent directionality of edges and attribute-edge correlations when dealing with decentralized directed social graphs, leading to synthetic graphs with poor utility. To bridge this gap, we present PrivAGM, a new solution that harnesses the synergies among differential privacy, secure multiparty computation, and generative graph models, enabling the secure construction of differentially private directed attributed graph models on decentralized social graphs while ensuring the privacy preservation of individuals. We evaluate PrivAGM on three real-world directed social graph datasets. The results show that PrivAGM outperforms the state-of-the-art methods, generating synthetic graphs with significantly higher utility.

1 INTRODUCTION

With the continuous advancements in graph analytics, a multitude of valuable applications can stem from the in-depth exploration of a social graph. However, social graph analytics becomes a challenging issue when the entire graph is not available to a single entity and stored in a *decentralized* manner. In this scenario, each user possesses only a limited local view comprising its attributes and neighbor list in the social graph, and the complete information of the social graph is formed by the collective views of all users. Decentralized social graphs have various practical applications [4, 20, 33, 36]. For example, the phone network exemplifies a decentralized social graph, where each user maintains their own profile (i.e., attributes) and contact list (i.e., neighbor list), and the combination of all users' profiles and contact lists constitutes the social graph [33]. Another concrete example is the decentralized social networks (e.g., diaspora* [8] and Mastodon [27]), where the complete social graph exists in a decentralized manner and no single server can access the entire social graph.

In the context of decentralized social graphs, the collection of individual users' local views for analytics can lead to significant privacy concerns, particularly when these local views contain sensitive information about social interactions, political preferences, and religious affiliations [4, 20, 21, 33, 36]. If users' local views are not sufficiently protected, it would discourage them from participating in such analytics. Thus, it is crucial to incorporate security measures into analytics over decentralized social graphs from the outset, facilitating the development and deployment of valuable applications without compromising users' privacy.

Privacy-preserving analytics over decentralized social graphs has received significant attention in recent years [3, 20–22, 33, 36, 44]. Most existing methods [20–22, 36] have adopted local differential privacy (LDP) mechanisms [24] to collect certain noisy statistics from the decentralized social graph for use. However, this line of work suffers from a major deficiency that limits their practical usability. Specifically, as more statistics are collected from the decentralized social graph, the total amount of revealed information monotonically increases, eventually reaching a threshold where collecting any new statistics leads to an excessive privacy risk [10].

Alternatively, a more promising direction is to privately fit a generative graph model to the decentralized social graph, producing a differentially private generative graph model. A synthetic graph is then generated from such model, which emulates the crucial characteristics of the decentralized graph and serves as a reliable proxy for conducting social graph analytics tasks. Based on the post-processing invariance property of differential privacy (DP) [12], the synthetic graph remains differentially private. Hay *et al.* [17] highlight three prominent advantages for utilizing synthetic graphs: (1) Certain analyses require running complex algorithms or simulations over actual social graph data, rather than relying solely on statistics. (2) Synthetic graphs facilitate "exploratory, open-ended, and iterative" analysis, which can be challenging to accommodate directly on an input private graph with limited privacy budget under DP. (3) It eliminates the need to share proprietary analysis details with the social graph owner. Based on these observations, this paper follows the direction of securely constructing differentially private generative graph models on decentralized social graphs.

Existing solutions. In the literature, there have been a number of works considering the setting of *centralized* graphs for the construction of differentially private generative graph models (e.g., [4, 23, 46], to list a few). On the contrary, little work [3, 33, 44, 48] has been done for constructing differentially private generative graph models over *decentralized* social graphs. The works in [3, 33, 48] concentrate solely on modeling graph structure, disregarding vertex

attributes and their correlations with the graph structure. Nevertheless, real-world social graphs possess vertex attributes and exhibit such correlations. For example, users often have various attributes (like age, gender, and sexual preference), and they are known to exhibit homophily, wherein users with similar attributes tend to form connections [32]. These correlations can be leveraged in various analyses; for instance, in the field of relational machine learning, they are used to predict missing or future attribute values [23]. The prior work most closely related to ours is AsgLDP [44], which aims to construct differentially private *attributed* graph models (AGMs) on decentralized social graphs.

However, AsgLDP, as well as other works [3, 33, 48], focuses on idealized decentralized *undirected* social graphs, where each user holds its *complete* neighbor list. In practice, however, decentralized social graphs are more commonly *directed*, where each user holds an *incomplete* neighbor list consisting of only "one-sided relationships". For instance, in phone networks, it is possible that Peter's contact list contains Tom's phone number, while Tom's contact list does not contain Peter's phone number. Another example is the Diaspora application [8], which incorporates the concept of "asymmetric following", enabling users to follow others without requiring mutual consent. Therefore, existing works are not effective in handling such a practical scenario. Additionally, all previous works do not consider the inherent directionality of the edges when constructing the generative graph model, leading to synthetic graphs with poor utility when dealing with directed graphs. Therefore, secure construction of differentially private *directed* AGMs (dAGMs) on decentralized directed social graphs remains to be fully explored.

Our contributions. In this paper, we design and evaluate PrivAGM, a new solution for secure construction of differentially private dAGMs on decentralized social graphs. PrivAGM is built from synergies among DP, secure multiparty computation (MPC), and generative graph models. We address three challenges when endeavoring to design PrivAGM.

Challenge 1: How to collect neighbor lists while striking a balance between privacy, utility, and efficiency?

To privately collect neighbor lists, a widely-used method is the randomized neighbor list (RNL) mechanism [33], which enables users to flip each bit in their neighbor lists with a certain probability to achieve pure edge LDP. However, the RNL mechanism significantly disrupts the structure of the decentralized social graph, as it requires flipping a considerable number of bits in each neighbor list to achieve the desired level of privacy protection. Another plausible method is to let users conceal each bit of their neighbor lists using MPC techniques (e.g., additive secret sharing [7]). However, this incurs significant performance overhead as the neighbor lists are normally sparse [6]. To strike a balance between privacy, utility, and efficiency, we instead propose layering the concept of selective MPC [19] with RNL to protect the neighbor lists. Unlike typical secret sharing-based MPC techniques, where all computing parties receive shares for all private values, selective MPC works by randomly selecting a subset of parties to receive shares for each private value. With our sparsity-aware neighbor list secret sharing method, PrivAGM achieves pure edge LDP for the neighbor lists *without introducing any noise that would compromise their utility* or incurring significant performance overhead.

Challenge 2: How to securely extract the differentially private characteristics essential for dAGM from the collected secret-shared social graph while ensuring their utility?

Constructing the differentially private dAGM requires the addition of calibrated noises to the extracted characteristics from the original graph. However, for certain characteristics (e.g., triangle subgraph count), the global sensitivity of the calibrated noises is large, which could severely compromise the utility of the extracted characteristics. To overcome this challenge, we develop techniques for oblivious edge clipping, enabling the oblivious restriction of the maximum degree of the secret-shared social graph. This helps mitigate the worst-case impact of a single vertex or edge and achieve a small global sensitivity. Moreover, we develop a suite of MPC-based secure components, including secure degree sequence extraction, secure attribute-edge correlation distribution extraction, secure attribute distribution extraction, and secure triangle counts extraction. Their integration allows to securely produce the differentially private characteristics essential for constructing the dAGM.

Challenge 3: How to construct the dAGM to effectively capture the inherent directionality of the edges in the decentralized social graph?

The existing AGMs [4, 23, 32, 44] have mainly focused on *undirected* graphs, resulting in a lack of capability to capture the inherent directionality of edges within the original graph. This directionality is of utmost importance for various analytics tasks on directed graphs, such as the discovery of social roles [11] and prediction of disease outbreaks [23]. However, no existing AGMs can be trivially adapted to capture the inherent directionality of edges. To address this, we identify key graph characteristics—such as vertex degrees, attribute-edge correlations, and triangle subgraph counts—that can capture the directionality, filling the gap in the field of AGMs. Specifically, we begin with the widely used Chung-Lu (CL) model [5], which operates on undirected graphs, and develop a directed version of the CL model to incorporate edge directionality as reflected in vertex degrees. Building on this directed CL model, we modify the classical AGM framework [23] to capture the directionality in attribute-edge correlations and triangle subgraph counts. This results in the dAGM, a model that generates directed attributed graphs, capturing the edge directionality of the original graph.

Evaluation results. We conduct experiments on three real-world social graph datasets. The results clearly demonstrate that, in terms of utility of synthetic graphs, PrivAGM not only outperforms the state-of-the-art decentralized setting-based method AsgLDP [44] but also surpasses the state-of-the-art centralized setting-based method CAGM [4]. For instance, in terms of the Hellinger distance between the degree distribution of the synthetic and input social graphs, PrivAGM is 58.3% lower than AsgLDP, and also is 42.9% lower than CAGM, with the privacy budget $\varepsilon = 1$ on the Gplus dataset [28]. We also evaluate the system cost. Notably, with $\varepsilon = 1.2$, PrivAGM's sparsity-aware neighbor list secret sharing method achieves about 70% savings in the size of resulting shares over the simple method of secretly sharing every bit of the neighbor lists.

2 PRELIMINARIES

2.1 Attributed Graphs

This paper focuses on directed and unweighted attributed graphs. An attributed graph is represented as $G = (\mathbf{A}, \mathbf{F})$. Here, $\mathbf{A} \in$

$\{0,1\}^{N\times N}$ is the asymmetric binary adjacency matrix of size $N\times N$, where N is the number of vertices. Each vertex is denoted by $i \in [1, N]$. $\mathbf{A}[i,j] = 1$ indicates the presence of a directed edge from vertex i to vertex j, while $\mathbf{A}[i,j] = 0$ indicates the absence of such a directed edge. We exclude self-connected edges in our consideration, meaning that $\mathbf{A}[i,i] = 0, i \in [1, N]$. The number of edges in G is denoted as M. We use $\mathbf{A}[i,:]$ to denote the i-th row of $\mathbf{A}$, which corresponds to the out-neighbors[1] of vertex i. The number of vertex i's out-neighbors (resp. in-neighbors) is named as its out-degree (resp. in-degree), denoted as d_i^+ (resp. d_i^-). $\mathbf{F} \in \{0,1\}^{N\times L}$ is the attribute matrix, where L is the dimensions of each vertex's attribute vector. $\mathbf{F}[i,:]$ is vertex i's attribute vector. The attribute vectors are assumed to be binary, following previous works [4, 23, 32, 44].

We use $\{\sigma_i\}_{i\in[1,\gamma]}$ to denote the set $\{\sigma_1, \cdots, \sigma_\gamma\}$, and omit the subscript $i \in [1,\gamma]$ when it does not impact the presentation.

2.2 Attributed Graph Model (AGM)

We provide an overview of the classical and widely adopted AGM [32] as a foundational approach to demonstrate AGM's underlying principles. Given the input graph, the AGM learns:

- The degree sequence of all vertices: $\mathcal{D} = \{d_i\}_{i\in[1,N]}$.
- The attribute distribution $\Theta_F : \Theta_F(\mathbf{y}) = \frac{n_\mathbf{y}}{N}$, where $n_\mathbf{y}$ is the number of vertices in the graph that have attribute vector $\mathbf{y}$.
- The attribute-edge correlation distribution $\Theta_X : \Theta_X(\mathbf{y}-\mathbf{y}') = \frac{n_{\mathbf{y}-\mathbf{y}'}}{M}$, where $\mathbf{y}-\mathbf{y}'$ is named as *attribute-edge correlation* (i.e., an edge connecting two vertices with attribute vector $\mathbf{y}$ and $\mathbf{y}'$, respectively), $n_{\mathbf{y}-\mathbf{y}'}$ is the number of $\mathbf{y}-\mathbf{y}'$ in the input graph.

To generate a graph, the AGM samples an attribute vector for each vertex from Θ_F. It then iteratively samples edges based on the new attributes from $\mathcal{D}$ and Θ_X.

2.3 Differential Privacy for Graphs

In this paper, we consider two variants of DP in the context of attributed graphs: edge LDP [33] and edge DP [23].

Definition 1. *(Edge LDP [33]) A randomized mechanism $\mathcal{M}$ satisfies ε-edge LDP, if and only if for any two neighbor lists $\boldsymbol{A}[i,:], \boldsymbol{A}'[i,:]$ that differ in one bit, we have: $\forall \boldsymbol{b} \in Range(\mathcal{M})$,*

$$Pr[\mathcal{M}(\boldsymbol{A}[i,:]) = \boldsymbol{b}] \le e^{\varepsilon} \cdot Pr[\mathcal{M}(\boldsymbol{A}'[i,:]) = \boldsymbol{b}],$$

where $Range(\mathcal{M})$ denotes the set of all possible outputs of $\mathcal{M}$, ε is the privacy budget.

Definition 2. *(Edge DP [23]) A randomized mechanism $\mathcal{M}$ satisfies (ε,δ)-edge DP, if and only if for any two neighboring attributed graphs G and G' that differ in the presence of a single edge or in the attribute associated with a single vertex, we have: $\forall \mathcal{G} \in Range(\mathcal{M})$,*

$$Pr[\mathcal{M}(G) = \mathcal{G}] \le e^{\varepsilon} \cdot Pr[\mathcal{M}(G') = \mathcal{G}] + \delta,$$

where δ is privacy parameter. If $\delta = 0$, we say that $\mathcal{M}$ provides pure edge DP. Note that unlike edge DP in the context of unattributed graphs [16], which exclusively addresses edge privacy, edge DP in the context of attributed graphs [23] goes a step further to protect the privacy of vertex attributes.

[1]Vertex i's out-neighbors are the vertices reachable by following its outgoing edges, while its in-neighbors are the vertices with directed edges pointing towards it.

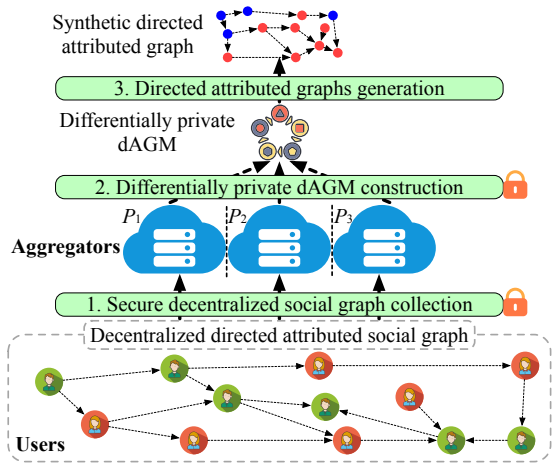


Figure 1: The system architecture of PrivAGM.

The discrete Laplace distribution $Lap(\cdot)$ is commonly employed to draw discrete noises to provide DP.

Definition 3. *(Discrete Laplace distribution [18]) A discrete random variable follows $Lap(\varepsilon,\delta,\Delta)$ if its probability mass function:*

$$Pr[x] = \frac{e^{\frac{\varepsilon}{\Delta}} - 1}{e^{\frac{\varepsilon}{\Delta}} + 1} \cdot e^{\frac{-\varepsilon\cdot|x-\mu|}{\Delta}}. \tag{1}$$

where μ is the mean of the distribution and Δ is the sensitivity.

2.4 Additive Secret Sharing

In 2-out-of-2 additive secret sharing (ASS) [7], a secret value $x \in \mathbb{Z}_{2^l}$ is split into two secret shares $\langle x\rangle_1$ and $\langle x\rangle_2$, which are distributed to two parties P_1 and P_2, respectively. We denote the ASS of x as $[\![x]\!]$. If only certain values in an entity Ω (e.g., set or table) are secret-shared, we represent the entity as $[\![\Omega]\!]$. The basic secure operations in the ASS domain are as follows: (1) Linear operations: Linear operations on secret-shared values can be computed locally. Given constants c, c' and secret-shared values $[\![x]\!], [\![y]\!]$, computing $[\![z]\!] = [\![c\cdot x + c'\cdot y]\!]$ involves $P_{i\in\{1,2\}}$ locally computing $\langle z\rangle_i = c\cdot\langle x\rangle_i + c'\cdot\langle y\rangle_i$. (2) Multiplication operations: Multiplication operation on two secret-shared values $[\![x]\!], [\![y]\!]$ requires one round of online communication. To compute $[\![z]\!] = [\![x\cdot y]\!]$, $P_{1,2}$ prepare a secret-shared Beaver triple $([\![w]\!], [\![u]\!], [\![v]\!])$ offline, where $w = u\cdot v$. $P_{i\in\{1,2\}}$ first computes $\langle e\rangle_i = \langle x\rangle_i - \langle u\rangle_i$ and $\langle f\rangle_i = \langle y\rangle_i - \langle v\rangle_i$, then communicates with each other to reveal e and f. Finally, $P_{1,2}$ compute $\langle z\rangle_1 = e\cdot f + f\cdot\langle u\rangle_1 + e\cdot\langle v\rangle_1 + \langle w\rangle_1$ and $\langle z\rangle_2 = f\cdot\langle u\rangle_2 + e\cdot\langle v\rangle_2 + \langle w\rangle_2$, respectively.

3 PROBLEM STATEMENT

3.1 System Architecture

PrivAGM's system architecture is illustrated in Fig. 1. In the system, there are two types of entities: users and aggregators. The N users, along with the connections among them, form the decentralized social graph $G = (\mathbf{A}, \mathbf{F})$. Each user corresponds to a vertex $i \in [1, N]$ and holds a local view $(\mathbf{A}[i,:], \mathbf{F}[i,:])$, where $\mathbf{A}[i,:]$ represents user i's neighbor list held by it (named as *local neighbor list*) and $\mathbf{F}[i,:]$ represents user i's attribute vectors. In contrast to previous studies [33, 44] that focus on idealized decentralized social

graphs where each user's local neighbor list *completely* includes its out/in-neighbors, i.e., $\mathbf{A}$ is symmetric, our attention is directed towards a more realistic *directed* social graph scenario. Here, each user's local neighbor list only includes its out-neighbors[2], i.e., $\mathbf{A}$ is asymmetric. Such decentralized social graphs are more commonly encountered in practice (e.g., the phone networks [33]).

The aggregators aim to collect the decentralized social graph data from the users to perform various graph analytics tasks. However, privacy concerns hinder users' participation in the tasks [33, 46]. Our goal is to facilitate secure construction of a dAGM on the decentralized social graph by the aggregators, while preserving the privacy of individuals. In this paper, we leverage a distributed trust framework where three aggregators (referred to as P_1, P_2, and P_3) from different trust domains cooperatively construct the dAGM. In practice, $P_{\{1,2,3\}}$ could be cloud servers hosted by different competitive commercial cloud providers (e.g., Amazon, Microsoft, and Google) or managed by distinct and potentially competing organizations. Such framework has recently gained significant attentions in both academia [35, 37–39, 41, 47] and industry [15, 29, 30].

3.2 Threat Model and Security Guarantees

Threat model. We consider a semi-honest and non-colluding adversary model, where each aggregator honestly follows the protocol, but may *individually* infer users' local views during the execution of PrivAGM. Such non-colluding multi-server model has been widely adopted for building secure database applications (e.g., [35, 39, 41, 47]). Additionally, we assume that the users are trusted, as they only provide their local views.

Security guarantees. From the view of individual aggregators, PrivAGM guarantees the following: (1) during the decentralized social graph collection phase, the local neighbor lists are differentially private (edge LDP); (2) during the dAGM construction phase, the learned dAGM is differentially private (edge DP). Based on the post-processing invariance of DP [12], the synthetic graphs generated by the differentially private dAGM are still differentially private.

4 SECURE DECENTRALIZED SOCIAL GRAPH COLLECTION

We first introduce how each user $i \in [1, N]$ secretly shares its neighbor list $\mathbf{A}[i,:]$ among $P_{\{1,2,3\}}$.

Motivation. To protect $\mathbf{A}[i,:]$, a simple method is to directly apply lightweight ASS over the entire $\mathbf{A}[i,:]$ of length N (i.e., dense encoding). However, this method is inefficient due to the sparsity of social graphs. According to Facebook's statistics [6], the average user has around 130 friends in a social network, which is much less than the total number of vertices, i.e., N. As a result, $\mathbf{A}[i,:]$ will be mostly filled with zeros, leading to high sparsity. Therefore, applying ASS over $\mathbf{A}[i,:]$ would result in significant uplink communication cost from user i to the aggregators, as well as unnecessary workload during the subsequent dAGM construction on the secret-shared social graph. Another simple method is to consider only the non-zero bits in $\mathbf{A}[i,:]$ (i.e., sparse encoding) and apply ASS solely to the indices of these non-zero elements. Specifically, user i first stores $\mathbf{A}[i,:]$ in sparse representation: $E_i = \{(i,j) | \mathbf{A}[i,j] = 1, j \in [1,N]\}$, and then secretly shares E_i as $[E_i] = \{(i, [\![j]\!]) | \mathbf{A}[i,j] = 1, j \in [1,N]\}$ ($[E_i] = \{(i, [\![j]\!])\}$ for simplicity). However, the secret sharing of indices makes it difficult to securely access the neighbors of each user. Since accessing neighbors is a fundamental operation required to extract parameters necessary for dAGM, the method also imposes a heavy workload on the aggregators.

Observation. We observe that while both offering privacy protection, the two aforementioned methods represent two extremes: the first one allows efficient accessing of neighbors but sacrifices sparsity, while the second one preserves sparsity but hinders efficient neighbor accessing. The observation indicates that the challenge here lies in balancing the preservation of privacy and the benefits of sparsity, while facilitating efficient neighbor accessing. We note that employing edge LDP [33] is a strategic approach to address the challenge. To achieve edge LDP, our starting point is the randomized neighbor list (RNL) mechanism. RNL applies randomized response [42], allowing each user to flip each bit of its local neighbor list with a certain probability. However, to achieve the desired level of privacy protection, RNL must flip a considerable number of bits in each local neighbor list, which significantly disrupts the structure of the decentralized social graph. To address this issue, our insight is to layer the concept of selective MPC [19] with RNL to achieve pure edge LDP for the local neighbor lists *without compromising their utility* or incurring significant performance overhead. Unlike typical secret sharing-based MPC techniques [7], where all computing participants receive secret shares for all private values, selective MPC randomly selects a subset of participants to receive secret shares for each private value.

Layering selective MPC with RNL. Given $\mathbf{A}[i,:]$, user i first constructs the set $E_i = \{(i,j,1) | \mathbf{A}[i,j] = 1, j \in [1,N]\}$ to fully capture the nonzero bits in $\mathbf{A}[i,:]$. Next, for every zero bit $\mathbf{A}[i,j] = 0, j \in [1,N]$, user i includes $(i,j,0)$ to E_i *with a probability of* p. The resulting local neighbor list is denoted as $E_i = \{(i,j,b_{ij})\}$, where $b_{ij} = 1$ or 0. User i then applies ASS over each b_{ij} in E_i to obtain the secret shares $[E_i] = \{(i,j,[\![b_{ij}]\!])\}$. After that, a straightforward method is to distribute $\{(i,j,\langle b\rangle_1)\}$ and $\{(i,j,\langle b\rangle_2)\}$ to P_1 and P_2, respectively. However, this approach cannot achieve pure edge LDP in the individual view of P_1 or P_2 because it solely *adds* zero bits to E_i and does not *remove* nonzero bits from E_i, thereby capturing only additive noise without accounting for subtractive noise. Our insight to address this issue is to leverage the concept of selective MPC [19]. Specifically, for *each* $(i,j,[\![b_{ij}]\!]) \in [E_i]$, user i sends $(i,j,\langle b\rangle_1)$ and $(i,j,\langle b\rangle_2)$ to two *randomly* selected aggregators, respectively. Thus, each aggregator only has access to a random subset of $[E_i]$. In other words, our method simulates removing some nonzero bits from the view of each aggregator, capturing subtractive noise (similar to how adding zero bits captures additive noise), thereby achieving pure edge LDP in each aggregator's view (as proven in Section 7.2) without compromising the utility.

For each $(i,j,[\![b_{ij}]\!]) \in [E_i]$, user i allows the two selected aggregators to be aware of each other to facilitate subsequent computations in the ASS domain. We use $\{[E_i]^{(1,2)}\}$, $\{[E_i]^{(2,3)}\}$, and $\{[E_i]^{(3,1)}\}$ to represent the secret-shared local neighbor lists of all users jointly held by the pairs of aggregators (P_1,P_2), (P_2,P_3), and (P_3,P_1), respectively, and use $\{[E_i]\} = \{[E_i]^{(1,2)} \cup [E_i]^{(2,3)} \cup [E_i]^{(3,1)}\}$ to represent the complete secret-shared neighbor lists.

[2]PrivAGM can be effortlessly expanded to the scenario where each user holds its in-neighbors.

Secretly sharing the attributes. Before encrypting the attribute vector $\mathbf{F}[i,:]$, user i first parses it into a value:

$$f_i = \sum_{l \in [1,L]} \mathbf{F}[i,l] \cdot 2^l. \tag{2}$$

User i secretly shares f_i into $[\![f_i]\!]$ and distributes $\langle f_i \rangle_1$ to P_1; $\langle f_i \rangle_2$ to P_2. Finally, the secret-shared graph is $(\{[E_i]\}, \{[\![f_i]\!]\})$.

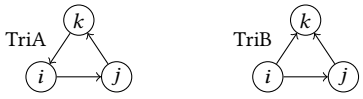
5 PRIVATE DAGM CONSTRUCTION

Next, we describe how the aggregators securely construct a differentially private dAGM on the collected secret-shared graph. Note that during the graph collection phase, users do not add any true noise to their local views. As a result, if the aggregators directly recover the dAGM parameters extracted from the secret-shared graph, the resulting dAGM will not be differentially private. Therefore, during the dAGM construction phase, the aggregators should add secret-shared noise to the dAGM parameters before recovering them, ensuring the construction of a differentially private dAGM.

Parameters necessary for dAGM. Our dAGM is built on the widely adopted AGM [32], which involves learning three sets of parameters from the input attributed graph: the degree sequence $\mathcal{D}$, the attribute-edge correlation distribution Θ_X, and the attribute distribution Θ_F (as introduced in Section 2.2). However, the AGM [32] focuses on undirected graphs, whereas PrivAGM specifically addresses directed graphs, where the *inherent directionality* of edges plays a crucial role in mimicking the original graph. Therefore, we extend the AGM [32] by adapting its required parameters, enabling to effectively capture the inherent directionality present in the input directed graph. Since the attribute distribution does not possess directionality, our focus lies solely on adapting the degree sequence and attribute-edge correlation distribution.

When considering the degree sequence, it is important to note that in directed graphs, each vertex has both out-degree and in-degree. Therefore, to effectively capture the directionality using the degree sequence, we tailor our dAGM to learn the sequence of (out-degree, in-degree) pairs, denoted as $\mathcal{D}^{\pm} = \{(d_i^+, d_i^-)\}_{i \in [1,N]}$. When considering the attribute-edge correlation distribution, to effectively capture the directionality, we tailor our dAGM to learn $\Theta_{\vec{X}} : \Theta_{\vec{X}}(f \to f') = \frac{n_{f \to f'}}{M}, f \to f' \in \mathbb{Z}_{2^L} \times \mathbb{Z}_{2^L}$, where $n_{f \to f'}$ is the number of edges *from* a vertex with attribute value f (mapped from an attribute vector by Eq. 2) *to* a vertex with attribute value f'. The distinction between our $\Theta_{\vec{X}}$ and the AGM [32]'s Θ_X lies in our consideration of attribute-edge correlations $f \to f'$ and $f' \to f$ as distinct, whereas the AGM [32] treats them as equivalent. As a result, our $\Theta_{\vec{X}}$ effectively captures the edge directionality.

The work [23] points out that the count of triangles in the input graph plays a pivotal role in refining the structure of synthetic graphs. However, it considers all triangles to be identical, thus also failing to capture the directionality of edges. To address the limitation, we propose counting non-isomorphic triangles separately. Specifically, we identify two non-isomorphic triangles "TriA" and "TriB", regardless of the vertex attributes:



Our dAGM counts the number of occurrences of TriA, TriB in the input graph separately, denoted as $n_{\triangle}^A, n_{\triangle}^B$, respectively.

Algorithm 1 Basic Oblivious Edge Clipping

Input: Secret-shared neighbor lists $\{[E_i]\}$; clipping parameter k.
Output: Secret-shared clipped local neighbor lists $\{[\overline{E}_i]\}$.

// $P_{1,2}$ operate on $\{[E_i]^{(1,2)}\}$:
Initialization: $[\![d_i^+]\!] = [\![0]\!]; [\![d_i^-]\!] = [\![0]\!]; [\overline{E}_i]^{(1,2)} = \emptyset, i \in [1,N]$.
for all $(i, j, [\![b_{ij}]\!]) \in [E_i]^{(1,2)}, i \in [1,N]$ **do**
  $[\![d_i^+]\!] = [\![d_i^+]\!] + [\![b_{ij}]\!]; [\![d_j^-]\!] = [\![d_j^-]\!] + [\![b_{ij}]\!]$.
  $[\![\mathsf{isClip}_1]\!] = \mathsf{secComp}([\![d_i^+]\!], k/3)$. // Oblivious comparison.
  $[\![\mathsf{isClip}_2]\!] = \mathsf{secComp}([\![d_j^-]\!], k/3)$. // Oblivious comparison.
  $[\![\mathsf{isClip}]\!] = [\![\mathsf{isClip}_1]\!] \cdot [\![\mathsf{isClip}_2]\!]$. // Aggregate the results.
  $[\![b'_{i,j}]\!] = [\![b_{ij}]\!] \cdot [\![\mathsf{isClip}]\!]$. // Obliviously clip the edge.
  $[\overline{E}_i]^{(1,2)}.\mathsf{add}((i, j, [\![b'_{i,j}]\!]))$. // Add the updated edge.
end for
$P_{2,3}$ (resp. $P_{3,1}$) perform the above operations on $\{[E_i]^{(2,3)}\}$ (resp. $\{[E_i]^{(3,1)}\}$) to produce $\{[\overline{E}_i]^{(2,3)}\}$ (resp. $\{[\overline{E}_i]^{(3,1)}\}$).
return $\{[\overline{E}_i]\} = \{[\overline{E}_i]^{(1,2)} \cup [\overline{E}_i]^{(2,3)} \cup [\overline{E}_i]^{(3,1)}\}$.

5.1 Oblivious Edge Clipping

Motivation. Edge clipping is necessary due to the significant sensitivity Δ involved in achieving edge DP for the attribute-edge correlation distribution $\Theta_{\vec{X}}$ and the triangle counts $n_{\triangle}^A, n_{\triangle}^B$. For example, the addition or removal of a single edge changes $n_{\triangle}^A$ or $n_{\triangle}^B$ by at most $d_{max} - 1$, where d_{max} is the maximum (out or in)-degree in the input graph, which can range up to $N - 1$. Setting such sensitivities and adding noises will completely destroy the utility of the extracted parameters. Hence, to achieve both high utility and edge DP in this scenario, it is crucial to mitigate the worst-case impact of a single vertex or edge. This can be accomplished by applying edge clipping on the original graph to restrict its maximum out/in-degree. A naive method would be for each user to locally perform edge clipping on their local neighbor lists. However, in the realistic decentralized social graph scenario, where each user holds an incomplete neighbor list, users cannot restrict their in-degrees.

Basic oblivious edge clipping. Our (basic) oblivious edge clipping algorithm is built on the widely-used edge clipping algorithm in the plaintext domain [1]. It projects an original graph with arbitrary degrees onto a graph where the maximum degree is k. Given a graph and a clipping parameter k, it iterates through each edge in a random order. If either of the two vertices connected by an edge currently has a degree larger than k, the edge is deleted.

However, the algorithm [1] focuses on undirected graphs, which means it does not differentiate between out/in-degrees. We adapt it to accommodate directed graphs. Specifically, given a directed graph, our algorithm iterates through its each directed edge in a random order. When considering an edge from vertex i to vertex j, their degrees d_i^+, d_i^- are incremented to 1. Then, if $d_i^+ > k$ or $d_i^- > k$, the edge is deleted. We then detail our secure protocol. As stated in Section 4, each aggregator only holds a subset of each local neighbor list, meaning that the (secret-shared) edges held by each aggregator are incomplete. Hence, they cannot simply clip the edges using the clipping parameter k. Our insight is to enable each pair of aggregators to *independently* perform oblivious edge clipping on their jointly held secret-shared local neighbor lists via a clipping

Algorithm 2 Efficient Oblivious Edge Clipping

Input: $\{[E_i]\}$; clipping parameters k; batch size w.
Output: Secret-shared clipped local neighbor lists $\{[\overline{E}_i]\}$.

//$P_{1,2}$ operate on $\{[E_i]^{(1,2)}\}$:
1: Initialization: $[\![d_i^+]\!] = [\![0]\!]; [\![d_i^-]\!] = [\![0]\!]; [\overline{E}_i]^{(1,2)} = \emptyset, i \in [1, N]$.
2: Restrict the out-degree of vertex $i \in [1, N]$ **in parallel**:
3: Initialize a queue $Q_i = \emptyset$.
4: **for all** $(i, j, [\![b_{ij}]\!]) \in [E_i]^{(1,2)}$ **do**
5: $\quad [\![d_i^+]\!] = [\![d_i^+]\!] + [\![b_{ij}]\!]; \quad Q_i.\text{push}((i, j, [\![b_{ij}]\!]))$.
6: $\quad$ **if** the size of the queue $Q_i \geq w$ **then**
7: $\quad\quad [\![\text{isClip}]\!] = \text{secComp}([\![d_i^+]\!], k/3)$.
8: $\quad\quad$ **while** $Q_i \neq \emptyset$ **do**
9: $\quad\quad\quad (i, j, [\![b_{ij}]\!]) = Q_i.\text{pop}()$.
10: $\quad\quad\quad [\![b'_{i,j}]\!] = [\![b_{ij}]\!] \cdot [\![\text{isClip}]\!]$. // Obliviously clip the edge.
11: $\quad\quad\quad [\overline{E}_i]^{(1,2)}.\text{add}((i, j, [\![b'_{i,j}]\!]))$. // Add updated edge to set.
12: $\quad\quad$ **end while**
13: $\quad$ **end if**
14: **end for**
15: Restrict the in-degrees on $[\overline{E}_i]^{(1,2)}$ **in parallel** as above.
16: $P_{2,3}$ (resp. $P_{3,1}$) perform the above operations on $\{[E_i]^{(2,3)}\}$ (resp. $\{[E_i]^{(3,1)}\}$) to produce $\{[\overline{E}_i]^{(2,3)}\}$ (resp. $\{[\overline{E}_i]^{(3,1)}\}$).
17: **return** $\{[\overline{E}_i]\} = \{[\overline{E}_i]^{(1,2)} \cup [\overline{E}_i]^{(2,3)} \cup [\overline{E}_i]^{(3,1)}\}$.

parameter of $k/3$. It ensures that, after the edge clipping, the maximum out/in-degree in the (secret-shared) subgraph held by each pair of aggregators is restricted to $k/3$. Therefore, the maximum out/in-degree is limited to $k/3 + k/3 + k/3 = k$. Due to the uniform random distribution of edges among $P_{1,2,3}$, the method yields a similar outcome as when clipping edges on complete neighbor lists by parameter k. Algorithm 1 shows the protocol.

Note that $\text{secComp}(\cdot, \cdot)$ (lines 4 and 5) is the oblivious comparison operation. Specifically, $\text{secComp}([\![\alpha]\!], \beta)$ outputs $[\![1]\!]$ if $\alpha \leq \beta$ and $[\![0]\!]$ if $\alpha > \beta$. We employ the efficient function secret sharing (FSS) based distributed comparison function (DCF) [2] to instantiate it. At line 6, $\text{isClip} = 1$ means that $\text{isClip}_1 = 1$ and $\text{isClip}_2 = 1$, indicating that $d_i^+ \leq k/3$ and $d_j^- \leq k/3$, and thus the edge $(i, j, [\![b_{ij}]\!])$ does not need to be clipped. $\text{isClip} = 0$ indicates that either $\text{isClip}_1 = 0$ or $\text{isClip}_2 = 0$, meaning that either $d_i^+ > k/3$ or $d_j^- > k/3$, and thus the edge $(i, j, [\![b_{ij}]\!])$ needs to be deleted. Finally, $P_{1,2}$ execute the oblivious edge clipping operation (line 7). The design intuition is that if $\text{isClip} = 0$, b'_{ij} is assigned 0, i.e., the edge is clipped, while if $\text{isClip} = 1$, b'_{ij} is assigned b_{ij}, i.e., the edge remains unchanged. Likewise, $P_{2,3}$ (resp. $P_{3,1}$) perform above operations on $\{[E_i]^{(2,3)}\}$ (resp. $\{[E_i]^{(3,1)}\}$).

Efficient oblivious edge clipping. While Algorithm 1 can achieve oblivious edge clipping, it comes with a notable communication round complexity. Specifically, as Algorithm 1 iterates through each edge *sequentially*, its communication rounds grow linearly with the number of edges. To address this issue, we propose an efficient protocol based on Algorithm 1, as given in Algorithm 2. The enhancement comes from the independence of the out-degree (resp. in-degree) of each vertex from the out-degree (resp. in-degree) of other vertices. Hence, Algorithm 2 enables the aggregators to simultaneously restrict each vertex's out/in-degree *in parallel* (lines 2 and 15). Another performance improvement from our proposed *batch edge clipping* technique. We observe that the true edges in the secret-shared local neighbor lists are sparse. This means that directly clipping the edges in a batch after a single check of the current degree (incremented by the edges in the batch), will seldom lead to clipping true edges. Hence, to reduce the number of required oblivious comparison operations, we modify the procedure by enabling the aggregators to check whether the current secret-shared out-degree or in-degree is larger than $k/3$ after iterating w edges (i.e., a batch), instead of performing the check after iterating each edge. The edges in each batch are then obliviously updated without distinction based on the secret-shared comparison result.

Algorithm 3 Secure Degree Sequence Extraction

Input: $\{[\overline{E}_i]\}$; k; privacy budget $\varepsilon_{\mathcal{D}}$; privacy parameter $\delta_{\mathcal{D}}$.
Output: Differentially private degree sequence $\widetilde{\mathcal{D}}^{\pm}$.

// P_1, P_2, and P_3 locally count the secret-shared degrees:
1: Initialization: $[\![d_i^+]\!] = [\![0]\!]; [\![d_i^-]\!] = [\![0]\!], i \in [1, N]$.
2: $[\![d_i^+]\!] = [\![d_i^+]\!] + [\![b_{ij}]\!], [\![d_j^-]\!] = [\![d_j^-]\!] + [\![b_{ij}]\!], (i, j, [\![b_{ij}]\!]) \in \{[\overline{E}_i]\}$.
3: P_3 secret-shares its shares of degrees with $P_{1,2}$.
4: $P_{1,2}$ add the received shares from P_3 to their own shares to obtain $[\![d_i^+]\!], [\![d_i^-]\!], i \in [1, N]$ in 2-out-of-2 ASS.
5: $P_{1,2}$ arrange the secret-shared degrees into table $[\![\mathbf{T}]\!]$, where $[\![\mathbf{T}[i, :]]\!] = [\![d_i^+]\!] \| [\![d_i^-]\!] \| [\![\rho]\!], i \in [1, N]$ and $\rho = 1$.
// P_3 constructs dummy degree pair table $\mathbf{T}'$:
6: $\mathbf{T}' = []$. // Empty table.
7: **for all** $(d^+, d^-), d^+ \in [0, k], d^- \in [0, k]$ **do**
8: $\quad n = \max(0, Lap(\varepsilon_{\mathcal{D}}, \delta_{\mathcal{D}}, 4))$. // Draw a noise.
9: $\quad$ Append n rows of $d^+ \| d^- \| \rho$ to $\mathbf{T}'$, where $\rho = 0$.
10: **end for**
11: P_3 secret-shares $\mathbf{T}'$ with $P_{1,2}$ to produce $[\![\mathbf{T}']\!]$.
// $P_{1,2}$ perform the remaining operations:
12: $[\![\hat{\mathbf{T}}]\!] = \frac{[\![\mathbf{T}]\!]}{[\![\mathbf{T}']\!]}$. // Vertical concatenation.
13: $[\![\hat{\mathbf{T}}']\!] = \text{secShuffle}([\![\hat{\mathbf{T}}]\!])$. // Oblivious row-wise shuffle.
14: Safely reveal columns 1 and 2 of $[\![\hat{\mathbf{T}}']\!]$ to obtain $[\hat{\mathbf{T}}']$.
15: Initialize degree histogram $[\![\mathbf{H}]\!] = [\![0]\!]^{(k+1)\times(k+1)}$.
16: **for all** $d^+ \| d^- \| [\![\rho]\!]$ in $[\hat{\mathbf{T}}']$ **do**
17: $\quad [\![\mathbf{H}[d^+, d^-]]\!] = [\![\mathbf{H}[d^+, d^-]]\!] + [\![\rho]\!]$. // Index edge to histogram.
18: **end for**
19: $[\![\widetilde{\mathbf{H}}[d^+, d^-]]\!] = [\![\mathbf{H}[d^+, d^-]]\!] + [\![Lap(\varepsilon_{\mathcal{D}}, 0, 4)]\!], d^+, d^- \in [0, k]$.
20: Safely reveal the differentially private histogram $\widetilde{\mathbf{H}}$.
21: $\widetilde{\mathbf{H}}'[d^+, d^-] = \max(0, \widetilde{\mathbf{H}}[d^+, d^-]), d^+, d^- \in [0, k]$.
22: $\widetilde{\mathbf{H}}''[d^+, d^-] = \widetilde{\mathbf{H}}'[d^+, d^-] \cdot \frac{N}{\text{sum}(\widetilde{\mathbf{H}}')}, d^+, d^- \in [0, k]$.
23: Initialize degree sequence $\widetilde{\mathcal{D}}^{\pm} = \emptyset$.
24: $\forall d^+, d^- \in [0, k]$, add $\widetilde{\mathbf{H}}''[d^+, d^-]$ tuples (d^+, d^-) to $\widetilde{\mathcal{D}}^{\pm}$.
25: **return** Differentially private degree sequence $\widetilde{\mathcal{D}}^{\pm}$.

5.2 Secure Degree Sequence Extraction

Design rationale. Algorithm 3 presents how $P_{1,2,3}$ securely extract the noisy version of the degree sequence $\mathcal{D}^{\pm}$ from the secret-shared clipped local neighbor lists $\{[\overline{E}_i]\}$, while ensuring that the extracted noisy sequence (denoted as $\widetilde{\mathcal{D}}^{\pm}$) is differentially private (edge DP). A naive method is to simply add secret-shared discrete Laplace noise to each secret-shared out/in-degree, followed by recovering the noisy degrees to obtain $\widetilde{\mathcal{D}}^{\pm}$. However, this method introduces

a significant amount of noise, particularly in low-degree vertices, which are abundant in real social graphs [23]. To tackle this issue, our solution is based on the insight that the mapping between a specific vertex and its degree is unimportant for how dAGM uses the degree sequence. What we actually need is the sequence of degrees, and the order is unimportant. Hence, $P_{1,2,3}$ extract the differentially private *histogram* of (out-degree, in-degree) pairs and then derive a more precise $\widetilde{\mathcal{D}}^{\pm}$ from the histogram.

Secure degree histogram construction. $P_{1,2,3}$ first construct the secret-shared histogram $[\![\mathbf{H}]\!]$. We observe that the primary challenge in constructing $[\![\mathbf{H}]\!]$ lies in securely indexing each secret-shared degree pair $([\![d_i^+]\!], [\![d_i^-]\!])$ within $[\![\mathbf{H}]\!]$. To achieve this, our insight is to let $P_{1,2}$ obliviously shuffle the secret-shared degree pairs based on a random permutation unknown to them to break the mappings between the degree pairs and the vertices, and then securely reveal the shuffled degree pairs to index them in $[\![\mathbf{H}]\!]$.

However, simply recovering the secret-shared degree pairs after the oblivious shuffle will directly expose $\mathbf{H}$. To address this, we let P_3 secret-share a certain number of dummy entities $d^+ \| d^- \| \rho$ (with flag $\rho = 0$) for each possible degree pair (d^+, d^-) with $P_{1,2}$ (lines 6-11). $P_{1,2}$ then perform the oblivious row-wise shuffle [14] (denoted as $\mathsf{secShuffle}(\cdot)$) on the real degree pairs (with flag $[\![\rho]\!] = [\![1]\!]$) together with the dummy degree pairs (with flag $[\![\rho]\!] = [\![0]\!]$), followed by revealing all degree pairs (without revealing the flags; lines 12-14). The oblivious shuffle ensures that $P_{1,2}$ cannot distinguish between the dummy and real degree pairs.

To ensure the revealed frequency of degree pairs in the views of $P_{1,2}$ is differentially private, the number of dummy entities for each possible degree pair is drawn from the discrete Laplace distribution $Lap(\varepsilon_{\mathcal{D}}, \delta_{\mathcal{D}}, 4)$ and truncated to 0 (line 8). Here, the sensitivity Δ is set to 4, as the addition or removal of an edge changes the out/in-degrees of exactly two vertices, resulting in changes to the frequency of at most four degree pairs. We will prove in Section 7.2 that the revealed degree pairs are differentially private in the individual view of P_1, P_2, even if the noises drawn by P_3 may be truncated to 0. Note that we cannot use the revealed frequency of degree pairs as $\mathbf{H}$ due to P_3 knowing the number of dummy entities for each degree pair, i.e., the frequency is not differentially private in P_3's view. Instead, $P_{1,2}$ add the secret-shared flag of each degree pair to the corresponding element of $[\![\mathbf{H}]\!]$ (lines 15-18).

Secure degree sequence construction. $P_{1,2}$ then construct $\widetilde{\mathcal{D}}^{\pm}$ based on $[\![\mathbf{H}]\!]$. $P_{1,2}$ first add a secret-shared noise drawn from $Lap(\varepsilon_{\mathcal{D}}, 0, 4)$ to each element of $[\![\mathbf{H}]\!]$ (line 19). The secret-shared noises can be generated offline by the protocol described in Section 4 of [13]. $P_{1,2}$ reveal the differentially private histogram $[\![\widetilde{\mathbf{H}}]\!]$ to obtain $\widetilde{\mathbf{H}}$. $\widetilde{\mathcal{D}}^{\pm}$ can be constructed based on $\widetilde{\mathbf{H}}$ (lines 21-24).

5.3 Secure Attribute-Edge Correlation Distribution Extraction

We now present how $P_{1,2,3}$ securely extract the noisy version of attribute-edge correlation distribution $\Theta_{\vec{X}}$, while ensuring that the extracted noisy distribution (denoted as $\widetilde{\Theta}_{\vec{X}}$) is differentially private. Algorithm 4 gives the protocol. Its design intuition shares similarities with Algorithm 3. Hence, we omit its introduction. Note that the edges held by each aggregator are incomplete. In addressing this problem, out insight is to let a specific pair of aggregators (assumed to be $P_{1,2}$) to compute $\widetilde{\Theta}_{\vec{X}}$ on their jointly held (incomplete) local neighbor lists. The uniform random distribution of edges among the three aggregators allows this approach to produce a distribution similar to that based on complete local neighbor lists. Note that the sensitivity Δ is set to $2 \cdot k/3$ (lines 6 and 15), as modifying the attribute value of a single vertex can affect, at most, $k/3$ edges (which corresponds to the maximum degree in $[\overline{E}_i]^{(1,2)}$), resulting in changes to the frequency of $2 \cdot k/3$ attribute-edge correlations.

Algorithm 4 Secure Attribute-Edge Distribution Extraction

Input: Clipped local neighbor lists $\{[\overline{E}_i]\}$; secret-shared attributes $\{[\![f_i]\!]\}$; maximum degree k; privacy budget, parameter ε_X, δ_X.
Output: Differentially private attribute-edge distribution $\widetilde{\Theta}_{\vec{X}}$.

// $P_{1,2}$ operate on $\{[\overline{E}_i]^{(1,2)}\}$ and $\{[\![f_i]\!]\}$:
1: $[\![\mathbf{T}]\!] = []$; $\mathbf{T}' = []$. // Empty tables.
2: **for all** $(i, j, [\![b_{ij}]\!]) \in [\overline{E}_i]^{(1,2)}, i \in [1, N]$ **do**
3: Append row $[\![f_i]\!] \| [\![f_j]\!] \| [\![\rho]\!]$ to $[\![\mathbf{T}]\!]$, where $[\![\rho]\!] = [\![b_{ij}]\!]$.
4: **end for**
// P_3 constructs dummy attribute-edge correlation table $\mathbf{T}'$:
5: **for all** $f \to f' \in \mathbb{Z}_{2^L} \times \mathbb{Z}_{2^L}$ **do**
6: $n = \max(0, Lap(\varepsilon_X, \delta_X, 2 \cdot k/3))$. // Draw a noise.
7: Append n rows of $f \| f' \| \rho$ to $\mathbf{T}'$, where $\rho = 0$.
8: **end for**
9: P_3 secret-shares $\mathbf{T}'$ with $P_{1,2}$ to produce $[\![\mathbf{T}']\!]$.
// $P_{1,2}$ perform the remaining operations:
10: $[\![\hat{\mathbf{T}}]\!] = \frac{[\![\mathbf{T}]\!]}{[\![\mathbf{T}']\!]}$. // Vertical concatenation.
11: $[\![\hat{\mathbf{T}}']\!] = \mathsf{secShuffle}([\![\hat{\mathbf{T}}]\!])$. // Oblivious row-wise shuffle
12: Safely reveal columns 1 and 2 of $[\![\hat{\mathbf{T}}']\!]$ to obtain $[\hat{\mathbf{T}}']$.
13: Initialize secret-shared histogram $[\![\mathbf{H}]\!] = [\![0]\!]^{2^L \times 2^L}$.
14: **for all** row $f \| f' \| [\![\rho]\!]$ in $[\hat{\mathbf{T}}']$ **do**
15: $[\![\mathbf{H}[f, f']]\!] = [\![\mathbf{H}[f, f']]\!] + [\![\rho]\!]$. // Index correlation to $[\![\mathbf{H}]\!]$.
16: **end for**
17: $[\![\widetilde{\mathbf{H}}[f, f']]\!] += [\![Lap(\varepsilon_X, 0, 2 \cdot k/3)]\!], f \to f' \in \mathbb{Z}_{2^L} \times \mathbb{Z}_{2^L}$.
18: Safely reveal the differentially private histogram $\widetilde{\mathbf{H}}$.
19: $\widetilde{\mathbf{H}}'[f, f'] = \max(0, \widetilde{\mathbf{H}}[f, f']), f \to f' \in \mathbb{Z}_{2^L} \times \mathbb{Z}_{2^L}$. // Truncation.
20: $\widetilde{\mathbf{H}}''[f, f'] = \frac{\widetilde{\mathbf{H}}'[f, f']}{\mathsf{sum}(\widetilde{\mathbf{H}}')}, f \to f' \in \mathbb{Z}_{2^L} \times \mathbb{Z}_{2^L}$. // Normalization.
21: Initialize attribute-edge distribution $\widetilde{\Theta}_{\vec{X}} = \{0\}^{2^L \times 2^L}$.
22: $\widetilde{\Theta}_{\vec{X}}(f \to f') = \widetilde{\mathbf{H}}''[f, f'], f \to f' \in \mathbb{Z}_{2^L} \times \mathbb{Z}_{2^L}$.
23: **return** Differentially private attribute-edge distribution $\widetilde{\Theta}_{\vec{X}}$.

Appendix A of the full version [40] details how to extract the differentially private attribute distribution (denoted as $\widetilde{\Theta}_F$) from $\{[\![f_i]\!]\}$. The privacy budget and parameter are ε_F and δ_F, respectively. The design intuition is identical to that of Algorithm 4.

5.4 Secure Triangle Counts Extraction

Algorithm 5 presents how to extract the noisy version of triangle counts $n_{\triangle}^A, n_{\triangle}^B$, while ensuring that the extracted noisy counts (denoted as $\widetilde{n}_{\triangle}^A, \widetilde{n}_{\triangle}^B$) are differentially private. The idea is to iterate over every possible set of three connected edges in pairs, incrementing the secret-shared product of flags for the three edges to the triangle count. However, since multiplication in the secret sharing domain requires communication between the involved parties, the method

Algorithm 5 Secure Triangle Counts Extraction

Input: $\{[\overline{E}_i]\}$; privacy budget ε_T; maximum in/out-degree k.
Output: Differentially private triangle counts $\widetilde{n}^A_\triangle$ and $\widetilde{n}^B_\triangle$.

1: Initialize $[\![n^A_\triangle]\!] = [\![0]\!]$; $[\![n^B_\triangle]\!] = [\![0]\!]$ in 3-out-of-3 ASS.
// $P_{1,2}$ operate on $\{[\overline{E}_i]^{(1,2)}\}$:
2: **for all** $i : i \in [1, N]$ **in parallel do**
3: **for all** $j : j < i$ and $(i, j, [\![b_{ij}]\!]) \in [\overline{E}_i]^{(1,2)}$ **do**
4: **for all** $k : k < i$ and $(j, k, [\![b_{jk}]\!]) \in [\overline{E}_j]^{(1,2)}$ **do**
5: **if** $(k, i, [\![b_{ki}]\!]) \in [\overline{E}_k]^{(1,2)}$ **then**
6: $[\![n^A_\triangle]\!] = [\![n^A_\triangle]\!] + [\![b_{ij}]\!] \cdot [\![b_{jk}]\!] \cdot [\![b_{ki}]\!]$. // Update the count.
7: **end if**
8: **end for**
9: **end for**
10: **end for**
11: **for all** $i : i \in [1, N]$ **in parallel do**
12: **for all** $j, k : j \neq k, (i, j, [\![b_{ij}]\!]), (i, k, [\![b_{ik}]\!]) \in [\overline{E}_i]^{(1,2)}$ **do**
13: **if** $(j, k, [\![b_{jk}]\!]) \in [\overline{E}_j]^{(1,2)}$ **then**
14: $[\![n^B_\triangle]\!] = [\![n^B_\triangle]\!] + [\![b_{ij}]\!] \cdot [\![b_{ik}]\!] \cdot [\![b_{jk}]\!]$. // Update the count.
15: **end if**
16: **end for**
17: **end for**
18: $P_{2,3}$ (resp. $P_{3,1}$) count triangles on $\{[\overline{E}_i]^{(2,3)}\}$ (resp. $\{[\overline{E}_i]^{(3,1)}\}$) as above and aggregate the triangle counts to $[\![n^A_\triangle]\!]$ and $[\![n^B_\triangle]\!]$.
19: $[\![n^A_\triangle]\!] = 9 \cdot [\![n^A_\triangle]\!]$; $[\![n^B_\triangle]\!] = 9 \cdot [\![n^B_\triangle]\!]$.
20: $[\![\widetilde{n}^A_\triangle]\!] = [\![n^A_\triangle]\!] + [\![Lap(\varepsilon_T, 0, k-1)]\!]$. //Add secret-shared noise.
21: $[\![\widetilde{n}^B_\triangle]\!] = [\![n^B_\triangle]\!] + [\![Lap(\varepsilon_T, 0, k-1)]\!]$. //Add secret-shared noise.
22: Safely reveal the differentially private triangle counts $\widetilde{n}^A_\triangle, \widetilde{n}^B_\triangle$.
23: **return** Differentially private triangle counts $\widetilde{n}^A_\triangle$ and $\widetilde{n}^B_\triangle$.

can only count triangles whose edges are held by the *same* pair of aggregators. To address this issue, our solution is based on the observation: the probability of each triangle's edges being distributed to the same pair of aggregators is $3 \cdot \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{1}{3} = \frac{1}{9}$. Hence, PrivAGM enables each pair of aggregators to count triangles over the edges they jointly hold. The counts are then multiplied by 9 to obtain approximate triangle counts. The sensitivity is set to $k-1$ as adding or removing an edge change the triangle count by at most $k-1$.

Remark. We note that performing Algorithm 5 sequentially would entail a significant number of communication rounds. As the iterations in the two outermost loops (i.e., lines 2 and 11) are independent of each other, we can execute them in parallel. Moreover, we observe that the primary cost arises from the secure 3-input multiplication gates in lines 6 and 14. To speed up Algorithm 5, we use the technique in Section B.3.1 of [31] to reduce the communication cost of each secure 3-input multiplication gate from 2 rounds and 8 elements to 1 round and 6 elements. Additionally, we use a random sampling strategy to further reduce the cost. Specifically, instead of iterating over each vertex, we only iterate over η randomly sampled vertices, and then multiply the secret-shared triangle counts by $\frac{N}{\eta}$.

6 ATTRIBUTED GRAPHS GENERATION

We detail how to generate directed attributed graphs. Our algorithm is built on the classical AGM framework [23]. However, it focuses on undirected graphs. Hence, our contribution is to adapt it to cater for the requirements of generating directed graphs.

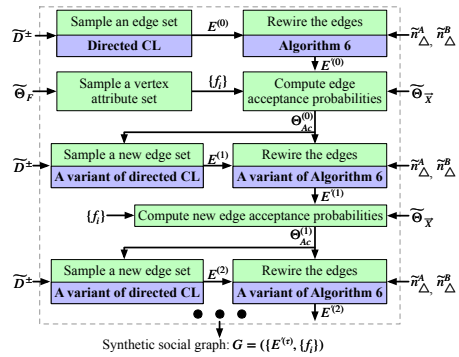


Figure 2: Overview of directed attributed graphs generation.

Overview. Fig. 2 gives an overview. We note that the CL model is a widely used generative graph model. However, it is designed exclusively for undirected graphs. Therefore, we propose a directed version of it, which is used to sample an initial *directed* edge set $E^{(0)} = \{(i, j)\}$ based on $\widetilde{\mathcal{D}}^{\pm}$. Next, Algorithm 6 is used to rewire the edges to ensure that the counts of TriA and TriB formed by $E^{(0)}$ are no less than $\widetilde{n}^A_\triangle$ and $\widetilde{n}^B_\triangle$, respectively. The rewired edges are denoted as $E'^{(0)}$. In addition, the vertex attribute set $\{f_i\}_{i\in[1,N]}$ (where f_i for vertex $i \in [1, N]$) is sampled from $\widetilde{\Theta}_F$. The accept/reject sampling technique is applied to generate new edge sets. Specifically, the edge acceptance probabilities $\Theta^{(0)}_{Ac}$ are computed based on the current graph $(E'^{(0)}, \{f_i\})$ and $\widetilde{\Theta}_{\vec{X}}$. Then a new edge set $E^{(1)}$ is sampled by the directed CL model and a variant of Algorithm 6 considering $\Theta^{(0)}_{Ac}$. This process iterates until Θ_{Ac} converges.

6.1 Directed CL Model

In the original CL model [5], each vertex is assigned a desired degree based on the degree sequence $\{d_i\}_{i\in[1,N]}$ of the input graph. Then, $\sum_{i\in[1,N]} d_i$ edges are sampled using a probability that is proportional to the desired degrees of the vertices. Specifically, an edge between vertices i, j is sampled with a probability of $\frac{d_i \cdot d_j}{\sum_{i\neq j, i,j\in[1,N]} d_i \cdot d_j}$. However, we observe that the model cannot capture the directionality of edges in the input *directed* graph. To overcome this limitation, we propose the *directed CL model* (denoted as directedCL$(\cdot)$).

In our directed CL model , we first assign each vertex i a desired out/in-degree pair (d^+_i, d^-_i) based on $\widetilde{\mathcal{D}}^{\pm}$. Then, for each directed edge from vertex i to vertex j, we sample it with a probability proportional to the out-degree of vertex i and the in-degree of vertex j, i.e., $\frac{d^+_i \cdot d^-_j}{\sum_{i\neq j, i,j\in[1,N]} d^+_i \cdot d^-_j}$. If the edge (i, j) is selected, (i, j) will be included in the edge set $E^{(0)}$. To maintain the expected degree distribution, we sample $\max(\sum_{i\in[1,N]} d^+_i, \sum_{i\in[1,N]} d^-_i)$ edges. Clearly, our model can capture the edge directionality in the original graph by sampling each edge based on the out-degree of the edge's starting vertex and the in-degree of the edge's ending vertex.

6.2 Refining Graph Structure

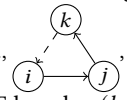
The (unattributed) graphs generated by the directed CL model only capture the degree information of the original graph, which

Algorithm 6 Refining Graph Structure with Edge Rewiring

Input: Edge set E; degree sequence $\widetilde{\mathcal{D}}^{\pm}$; triangle counts $\widetilde{n}^A_\triangle, \widetilde{n}^B_\triangle$.
Output: The new edge set E' after rewiring.

1: $n^A_\triangle, n^B_\triangle$ = countTriangle(E). // Count triangles on E.
2: **while** $n^A_\triangle < \widetilde{n}^A_\triangle$ and $n^B_\triangle < \widetilde{n}^B_\triangle$ **do**
3: // Construct TriA:
4: i = Sample($[1, N], \widetilde{\mathcal{D}}^{\pm}$). // Sample a vertex based on $\widetilde{\mathcal{D}}^{\pm}$.
5: j = SampleNbr(E, i). // Randomly sample a vertex from vertex i's out-neighbors.
6: k = SampleNbr(E, j).
7: **if** $(k, i) \notin E$ **then**
8: (q, r) = oldestEdge(E). // Get the oldest edge from E.
9: **if** replacing (q, r) by (k, i) boosts triangle counts **then**
10: E.replace($(q, r), (k, i)$). // Replace (q, r) by (k, i).
11: Make edge (k, i) the youngest edge in E.
12: $n^A_\triangle, n^B_\triangle$ = countTriangle(E).
13: **else**
14: Make edge (q, r) the youngest edge in E.
15: **end if**
16: **end if**
// Construct TriB:
17: i = Sample($[1, N], \widetilde{\mathcal{D}}^{\pm}$).
18: j, k = Sample2Nbr(E, i). // Randomly sample two vertices from vertex i's out-neighbors.
19: If $(j, k) \notin E$, replacing the oldest edge by (j, k) as above.
20: **end while**
21: **return** The edge set $E' = E$ after rewiring.

is insufficient to simulate the structure of the original graph. To address this limitation, the AGM [23] proposes an iterative edge rewiring method to ensure that the number of triangles in the generated graph is no less than that of the original graph, thereby enhancing the structural representation of the generated graph. However, when considering directed graphs, the AGM overlooks the inherent directionality of edges in the original graph and treats all triangles as identical. To overcome this limitation in the context of directed graphs, we propose Algorithm 6 (denoted as Rewire($\cdot$)).

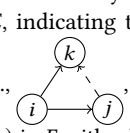
Algorithm 6 operates on the edge set E generated by the directed CL model. It iteratively rewires the edges until the counts of TriA and TriB formed by the edges surpass $\widetilde{n}^A_\triangle$ and $\widetilde{n}^B_\triangle$, respectively. Specifically, to create a new edge for constructing TriA (lines 4-16), the algorithm proceeds as follows: (1) sample a vertex i from $[1, N]$ where each vertex $i \in [1, N]$ is sampled with a probability of $\frac{d^+_i}{\sum_{i\in[1,N]} d^+_i}$; (2) randomly sample a vertex j from vertex i's out-neighbors; (3) randomly sample a vertex k from vertex j's out-neighbors. If $(k, i) \notin E$, indicating that vertices i, j, and k form an incomplete TriA, i.e., 

, the algorithm attempts to replace the oldest edge in E by edge (k, i) to construct TriA (lines 8-15). Specifically, if replacing the oldest edge (q, r) in E with edge (k, i) would increase both the counts of TriA and TriB, edge (q, r) is replaced by edge (k, i) and (k, i) is set as the youngest edge in E. Otherwise, edge (q, r) is set as the youngest edge in E.

Algorithm 7 Directed Attributed Graphs Generation

Input: The dAGM parameters $\widetilde{\mathcal{D}}^{\pm}$, $\widetilde{\Theta}_{\vec{X}}$, $\widetilde{\Theta}_F$, $\widetilde{n}^A_\triangle$, and $\widetilde{n}^B_\triangle$.
Output: The synthetic directed social graph $G = (E, \{f_i\})$.

1: $\{f_i\}_{i\in[1,N]}$ = Sample($\widetilde{\Theta}_F$). // Sample a vertex attribute set.
2: $E^{(0)}$ = directedCL($\widetilde{\mathcal{D}}^{\pm}$). // Sample the initial edge set.
3: $E'^{(0)}$ = Rewire($E^{(0)}, \widetilde{\mathcal{D}}^{\pm}, \widetilde{n}^A_\triangle, \widetilde{n}^B_\triangle$). // Rewire edges by Algo. 6.
4: Initialize the round $\tau = 0$.
5: **while** edge acceptance probability Θ_{Ac} does not converage **do**
6: Compute $\Theta^{(\tau)}_{\vec{X}}$ of the current graph $(E'^{(\tau)}, \{f_i\})$.
7: $R(f \to f') = \frac{\widetilde{\Theta}_{\vec{X}}(f\to f')}{\Theta^{(\tau)}_{\vec{X}}(f\to f')}, f \to f' \in \mathbb{Z}_{2^L} \times \mathbb{Z}_{2^L}$.
8: **if** $\tau > 0$ **then**
9: $R(f\to f') = R(f\to f') \cdot \Theta^{(\tau-1)}_{Ac}(f\to f'), f\to f' \in \mathbb{Z}_{2^L}\times\mathbb{Z}_{2^L}$.
10: **end if**
11: $\Theta^{(\tau)}_{Ac}(f \to f') = \frac{R(f\to f')}{\mathrm{SUP}(R)}, f \to f' \in \mathbb{Z}_{2^L} \times \mathbb{Z}_{2^L}$.
// Sample a new edge set $E'^{(\tau+1)}$ based on $\Theta^{(\tau)}_{Ac}$:
12: $E^{(\tau+1)}$ = directedCL($\widetilde{\mathcal{D}}^{\pm}, \{f_i\}, \Theta^{(\tau)}_{Ac}$).
13: $E'^{(\tau+1)}$ = Rewire($E^{(\tau+1)}, \widetilde{\mathcal{D}}^{\pm}, \widetilde{n}^A_\triangle, \widetilde{n}^B_\triangle, \{f_i\}, \Theta^{(\tau)}_{Ac}$).
14: $\tau = \tau + 1$.
15: **end while**
16: **return** $E = E'^{(\tau)}$; The synthetic social graph $G = (E, \{f_i\})$.

Similarly, Algorithm 6 creates a new edge to construct TriB as follows (lines 17-19). First, a vertex i is sampled from $[1, N]$ where each vertex $i \in [1, N]$ is chosen with a probability of $\frac{d^+_i}{\sum_{i\in[1,N]} d^+_i}$. Then, two vertices j and k are randomly sampled from vertex i's out-neighbors. If $(j, k) \notin E$, indicating that vertices i, j, and k form an incomplete TriB, i.e., 

, the algorithm checks if replacing the oldest edge (q, r) in E with edge (j, k) would increase both the counts of TriA and TriB. If it does, the algorithm replaces edge (q, r) with edge (j, k) and makes (j, k) the youngest edge in E. Otherwise, it makes edge (q, r) the youngest edge in E.

6.3 Putting It Together

Algorithm 7 gives the workflow for synthesizing directed graphs. Lines 6-11 calculate the new acceptance probabilities $\Theta^{(\tau)}_{Ac}$ based on the attribute-edge correlation $\Theta^{(\tau)}_{\vec{X}}$ in the current graph $(E'^{(\tau)}, \{f_i\})$ and $\widetilde{\Theta}_{\vec{X}}$ derived from the original graph. $\mathrm{SUP}(R)$ is the least upper bound of set R. The new edge set $E'^{(\tau+1)}$ is sampled (lines 12-13). Specifically, when the directed CL model or Algorithm 6 add an edge (i, j) to the new edge set, they first sample a value uniformly from the range of 0 to 1 (Sample$(0, 1)$). The edge (i, j) is included in the new edge set (i.e., "accepted") only when Sample$(0, 1) \leq \Theta^{(\tau+1)}_{Ac}(f_i \to f_j)$, where f_i and f_j are the attributes sampled for vertices i and j at line 1. If Sample$(0, 1) > \Theta^{(\tau+1)}_{Ac}(f_i \to f_j)$, the edge (i, j) is not included in the new edge set (i.e., "rejected").

7 PRIVACY AND SECURITY ANALYSIS

In Section 7.1, we parametrize the leakage function $\mathcal{L}$, which indicates the additional knowledge that the adversary may learn during

the execution of PrivAGM. In Section 7.2, we conduct a formal analysis to prove that the output of $\mathcal{L}$ is differentially private. In Section 7.3, we prove that the adversary's view can be simulated solely based on the output of $\mathcal{L}$.

7.1 Leakage Function

We define the leakage function as $\mathcal{L} = (\mathcal{L}_E, \mathcal{L}_{\mathcal{D}}, \mathcal{L}_X, \mathcal{L}_F, \mathcal{L}_\triangle)$:

• $\mathcal{L}_E = \{[E_i]\}$, where $[E_i] = \{(i, j, \langle b_{ij}\rangle)\}$ denotes the local neighbor list held by the adversary for user i.

• $\mathcal{L}_{\mathcal{D}} = ([\hat{\mathbf{T}}'_{\mathcal{D}}], \widetilde{\mathbf{H}}_{\mathcal{D}})$, where $[\hat{\mathbf{T}}'_{\mathcal{D}}]$ and $\widetilde{\mathbf{H}}_{\mathcal{D}}$ are the table and histogram revealed in lines 14 and 20 of Algorithm 3, respectively. While the adversary learns additional information during the execution of other operations in Algorithm 3, since the information is derived from $\widetilde{\mathbf{H}}_{\mathcal{D}}$, it is not included in $\mathcal{L}_{\mathcal{D}}$.

• $\mathcal{L}_X = ([\hat{\mathbf{T}}'_X], \widetilde{\mathbf{H}}_X)$, where $[\hat{\mathbf{T}}'_X], \widetilde{\mathbf{H}}_X$ are the table and histogram revealed in lines 12, 18 of Algorithm 4, respectively.

• $\mathcal{L}_F = ([\hat{\mathbf{T}}'_F], \widetilde{\mathbf{H}}_F)$, where $[\hat{\mathbf{T}}'_F]$ and $\widetilde{\mathbf{H}}_F$ are the table and histogram revealed in secure attribute distribution extraction.

• $\mathcal{L}_\triangle = (\widetilde{n}^A_\triangle, \widetilde{n}^B_\triangle)$, where $\widetilde{n}^A_\triangle$ and $\widetilde{n}^B_\triangle$ are the triangle counts revealed in line 22 of Algorithm 5.

7.2 Privacy Analysis

Theorem 1. *The neighbor list $[E_i]$ of user i held by the adversary satisfies pure ε_E-edge LDP with $\varepsilon_E = \max\{\ln(1/p), \ln(3-2p)\}$.*

We prove Theorem 1 in Appendix B of the full version [40]. We analyze in Appendix C of the full version how to reduce ε_E with additional aggregators.

Theorem 2. *Given $\mathcal{L}_{\mathcal{D}}$, in the view of adversary, $[\hat{\mathbf{T}}'_{\mathcal{D}}]$ adheres to $(\varepsilon_{\mathcal{D}}, \delta_{\mathcal{D}})$-edge DP and $\widetilde{\mathbf{H}}_{\mathcal{D}}$ adheres to pure $\varepsilon_{\mathcal{D}}$-edge DP.*

We prove Theorem 2 in Appendix D of the full version [40].

Theorem 3. *Given $\mathcal{L}_X$, in the view of adversary, $[\hat{\mathbf{T}}'_X]$ adheres to $(\varepsilon_X, \delta_X)$-edge DP and $\widetilde{\mathbf{H}}_X$ adheres to pure ε_X-edge DP.*

Proof. Since the proof follows the same idea as the proof of Theorem 2, we omit the proof. □

Theorem 4. *Given $\mathcal{L}_F$, in the view of the adversary, $[\hat{\mathbf{T}}'_F]$ adheres to $(\varepsilon_F, \delta_F)$-edge DP and $\widetilde{\mathbf{H}}_F$ adheres to ε_F-edge DP.*

Proof. Since the proof also follows the same idea as the proof of Theorem 2, we omit the proof. □

Theorem 5. *Given $\mathcal{L}_\triangle = (\widetilde{n}^A_\triangle, \widetilde{n}^B_\triangle)$, in the view of the adversary, both $\widetilde{n}^A_\triangle$ and $\widetilde{n}^B_\triangle$ adhere to pure $\varepsilon_\triangle$-edge DP.*

We prove Theorem 5 in Appendix E of the full version [40].

7.3 Security Analysis

We use the simulation paradigm [26] to analyze the security. We begin by introducing the ideal functionality $\mathcal{F}$:

(1) $\mathsf{Setup}(N, k, w)$: The aggregators send N, k, w to $\mathcal{F}$, where N is the number of users, k is the clipping parameter, w is the batch size. $\mathcal{F}$ executes the setup process with N, k, w.

(2) $\mathsf{Colle}(E_i, i \in [N])$: The users send their local neighbor lists $E_i, i \in [N]$ to $\mathcal{F}$.

(3) $\mathsf{dAGM}(\{E_i\}, N, k, w)$: $\mathcal{F}$ builds a dAGM on $\{E_i\}$, and then generates synthetic graphs via the dAGM.

We allow $\mathcal{F}$ to leak $\mathcal{L}^* = (\mathcal{L}, N, k, w)$.

Definition 4. *Let $\prod$ denote the protocol for securely constructing dAGM. Let $\mathcal{A}$ be an adversary who statically corrupts one of $P_{\{1,2,3\}}$ or one of the users, and let $\mathsf{View}^{\prod}_{\mathsf{Real}}$ be the view of the corrupted parties during the protocol run. In the ideal world, a simulator $\mathcal{S}$ generates a simulated view $\mathsf{View}^{\mathcal{S},\mathcal{L}^*}_{\mathsf{Ideal}}$ given only the leakage of $\mathcal{F}$. We say that $\prod$ is secure, if there exists a probabilistic polynomial time simulator $\mathcal{S}$ such that $\mathsf{View}^{\mathcal{S},\mathcal{L}^*}_{\mathsf{Ideal}}$ is indistinguishable from $\mathsf{View}^{\prod}_{\mathsf{Real}}$.*

Theorem 6. *Based on Definition 4, PrivAGM securely realizes $\mathcal{F}$ with the leakage $\mathcal{L}^*$.*

We prove Theorem 6 in Appendix F of our full version [40].

8 PERFORMANCE EVALUATION

8.1 Setup

We develop a prototype implementation of PrivAGM using a combination of Python and C++. All experiments are conducted on a workstation equipped with 24 Intel Xeon Gold 6240R CPU cores, a NVIDIA RTX A6000 GPU, 128 GB of RAM, and 2 TB of external SSD storage, running Ubuntu 20.04.3 LTS. We use a MacBook Air with 8 GB of RAM to simulate users, secretly share the local views. All public and private values are encoded in $\mathbb{Z}_{2^{32}}$. Given the overall privacy budget ε, we allocate $\varepsilon_{\mathcal{D}} = 0.6 \cdot \varepsilon, \varepsilon_X = 0.1 \cdot \varepsilon, \varepsilon_F = 0.1 \cdot \varepsilon, \varepsilon_\triangle = 0.1 \cdot \varepsilon$. In addition, we set the privacy parameters $\delta_{\mathcal{D}} = \delta_X = \delta_F = 10^{-5}$. The clipping parameter k follows the previous work [23] and is set to $\sqrt[3]{N}$. The clipping batch size w is set to $\sqrt{N}$.

Datasets. We use three directed social graph datasets: MOOC (7047 vertices; 411749 edges) [25], Twitter (81306 vertices; 1768149 edges) [28], and Gplus (107614 vertices; 13673453 edges) [28].

Baselines. We evaluate two state-of-the-art solutions as baselines for utility comparison in attributed graph synthesis: CAGM, which operates in a centralized setting with edge DP guarantees [4], and AsgLDP, which operates in a decentralized setting with edge LDP guarantees [44]. To ensure fairness, we use the recommended parameters from the original papers of the baselines[3].

Utility metrics. We evaluate the utility of the synthetic graphs using four widely-used metrics [4, 23, 33, 44]. (1) *Degree distribution*: We use the Kolmogorov-Smirnov statistic to evaluate how well a synthetic graph captures the degree distribution of the original graph (denoted as $\mathsf{KS}_{\mathsf{degree}}$). As $\mathsf{KS}_{\mathsf{degree}}$ is less sensitive to differences in the tails of the degree distributions, we also report the Hellinger distance (denoted as $\mathsf{HD}_{\mathsf{degree}}$) between the degree distributions of the synthetic and original graphs. (2) *Attribute-edge correlation distribution*: We report the mean relative error (MRE) (denoted as $\mathsf{MRE}_{\mathsf{corr}}$) and the Hellinger distance $\mathsf{HD}_{\mathsf{corr}}$ between the attribute-edge correlation distribution of the synthetic and original graphs. (3) *Local clustering coefficient*: We report the MRE between the average of the local clustering coefficient of the synthetic and original graphs (denoted as $\mathsf{MRE}_{\mathsf{cluster}}$). (4) *Triangle counts*: We report the MRE in the number of triangles, TriA and TriB (denoted

[3]We note that AsgLDP [44] has some technical flaws in its DP guarantee. However, to ensure a fair comparison, we continue to use its original protocol, as correcting the flaws would further degrade its performance.

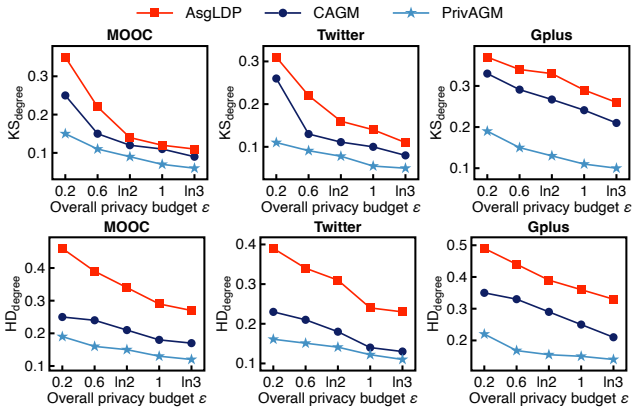


Figure 3: Comparison in terms of $\mathsf{KS}_{\mathsf{degree}}$ and $\mathsf{HD}_{\mathsf{degree}}$.

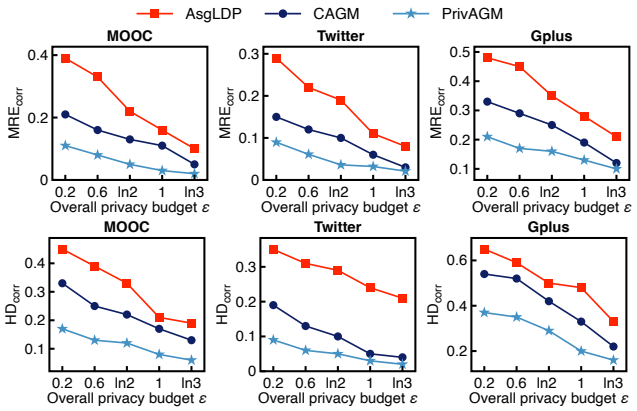


Figure 4: Comparison in terms of $\mathsf{MRE}_{\mathsf{corr}}$ and $\mathsf{HD}_{\mathsf{corr}}$.

as $\mathsf{MRE}_{\mathsf{TriA}}$ and $\mathsf{MRE}_{\mathsf{TriB}}$ respectively). Their computation method is given in Appendix G of the full version [40].

8.2 Utility Comparison

Degree distribution. Fig. 3 presents a comparison between PrivAGM and the baselines in terms of $\mathsf{KS}_{\mathsf{degree}}$ and $\mathsf{HD}_{\mathsf{degree}}$. It is evident that AsgLDP shows the largest $\mathsf{KS}_{\mathsf{degree}}$ and $\mathsf{HD}_{\mathsf{degree}}$ since it is based on LDP, which introduces larger noise than central DP under the same privacy budget. In contrast, our PrivAGM achieves the smallest $\mathsf{KS}_{\mathsf{degree}}$ and $\mathsf{HD}_{\mathsf{degree}}$. This is attributed to PrivAGM preserving the inherent edge directionality in the input graphs.

Attribute-edge correlation distribution. Fig. 4 shows a comparison between PrivAGM and the baselines in terms of $\mathsf{MRE}_{\mathsf{corr}}$ and $\mathsf{HD}_{\mathsf{corr}}$. PrivAGM consistently achieves the smallest MREcorr and HDcorr across all privacy budgets and datasets. Moreover, as ε decreases, the difference in MREcorr, HDcorr between PrivAGM and the baselines becomes more significant.

Local clustering coefficient. Fig. 5 presents a comparison between PrivAGM and the baselines in terms of $\mathsf{MRE}_{\mathsf{cluster}}$. We can observe that the $\mathsf{MRE}_{\mathsf{cluster}}$ of PrivAGM and CAGM is approximately similar. In certain cases (e.g., $\varepsilon = \ln 3$ on the Gplus dataset), the $\mathsf{MRE}_{\mathsf{cluster}}$ of CAGM is even lower than that of PrivAGM. This disparity is because $\mathsf{MRE}_{\mathsf{cluster}}$ fails to capture the difference in edge directionality between the synthetic and the original graphs.

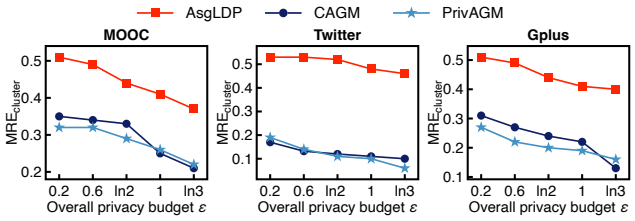


Figure 5: Comparison in terms of $\mathsf{MRE}_{\mathsf{cluster}}$.

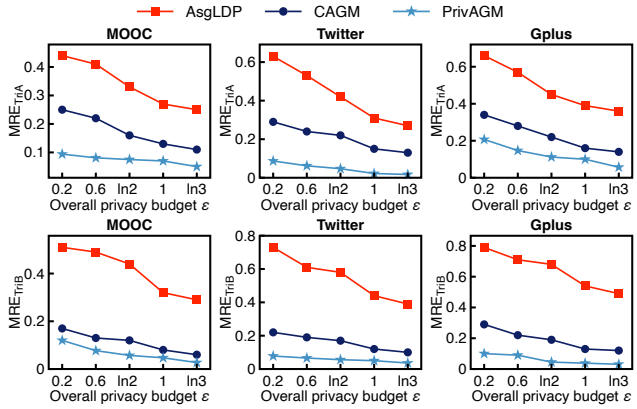


Figure 6: Comparison in terms of $\mathsf{MRE}_{\mathsf{TriA}}$ and $\mathsf{MRE}_{\mathsf{TriB}}$.

Hence, the advantage of PrivAGM in capturing the directionality of the original graphs cannot be fully reflected by $\mathsf{MRE}_{\mathsf{cluster}}$.

Triangle counts. Fig. 6 presents a comparison between PrivAGM and the baselines in terms of $\mathsf{MRE}_{\mathsf{TriA}}$ and $\mathsf{MRE}_{\mathsf{TriB}}$. The results demonstrate that PrivAGM consistently achieves a significantly small $\mathsf{MRE}_{\mathsf{TriA}}$, $\mathsf{MRE}_{\mathsf{TriB}}$ (even smaller than 0.1 in most instances) compared to the baselines and validate our sampling-based secure triangle counting scheme. In contrast, AsgLDP consistently exhibits the highest $\mathsf{MRE}_{\mathsf{TriA}}$ and $\mathsf{MRE}_{\mathsf{TriB}}$ (even exceeding 0.5 in most instances). This is because AsgLDP does not optimize the structure of the synthetic graphs by incorporating triangle counts. While CAGM optimizes the structure of synthetic graphs by triangle counts, it still cannot achieve lower $\mathsf{MRE}_{\mathsf{TriA}}$ and $\mathsf{MRE}_{\mathsf{TriB}}$ than PrivAGM.

8.3 System Cost Evaluation

We do not compare PrivAGM with the baselines in terms of system cost, as CAGM [4] considers a centralized setting and AsgLDP [44] has significant loss in the generated synthetic graphs.

Local neighbor list secret sharing. Fig. 7 presents the results of the lower bound ε_E for leakage in local neighbor list secret sharing and the savings on the resulting size. The results show that PrivAGM achieves a significant reduction in the resulting ciphertext size compared to direct secret sharing of the complete local neighbor list in dense encoding (up to 90% under $p = 0.1, \varepsilon_E \approx 2.3$). Moreover, the savings on the resulting ciphertext size remain consistent across different datasets, highlighting the scalability of our method. These savings will significantly reduce the system cost in the subsequent secure dAGM construction.

Oblivious edge clipping. Fig. 8 compares the basic and efficient oblivious edge clipping protocols. The results for the basic protocol

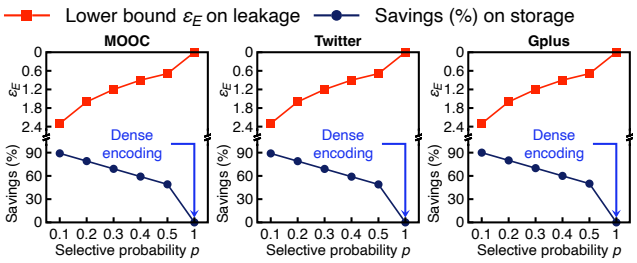


Figure 7: Evaluation of ε_E for leakage in neighbor list secret sharing and the savings on storage.

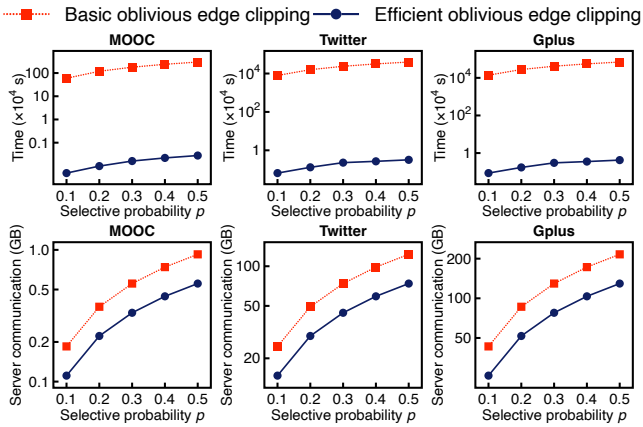


Figure 8: Comparison of oblivious edge clipping methods.

are estimated from processing 1000 edges, as it is significantly slow (taking a few days). The comparison shows that the efficient protocol achieves a substantial speedup compared to the basic protocol (up to 10000×). Additionally, the efficient protocol achieves approximately 50% savings in communication cost compared to the basic protocol. These performance improvements stem from the fact that the efficient protocol allows the aggregators to process the degree of each vertex in parallel and obliviously clip the edges in batches.

Differentially private dAGM construction. Fig. 9 shows the time and communication costs, along with their breakdown. We can observe that the majority of the cost can be attributed to the secure triangle counting. Its high time cost is primarily due to the communication required by the 3-input multiplication gates (i.e., lines 6, 14 of Algorithm 5). Additionally, the results further demonstrate that our local neighbor list secret sharing can significantly save the cost of secure dAGM construction. While the cost of dAGM construction is relatively high, it is a *one-off* process. Once the dAGM is constructed, it can generate numerous graphs.

9 RELATED WORK

Private centralized graph analytics: In the *centralized* setting, there exists a considerable body of work on developing methods for private graph analysis. In this setting, a curator possesses the complete graph and publishes some noisy information in a manner that satisfies DP. Some works publish differentially private statistics of the original graph, e.g., degree distribution [34] and subgraph

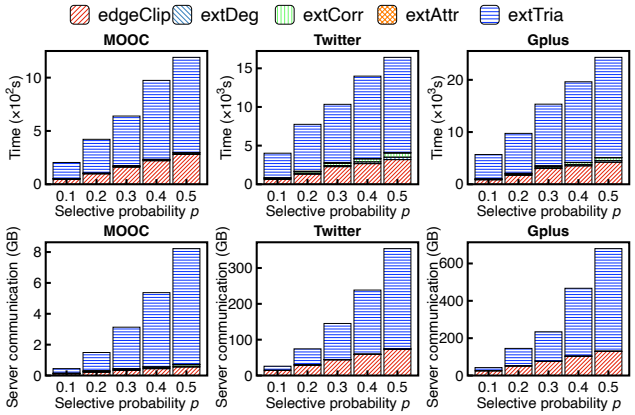


Figure 9: Time and communication cost of differentially private dAGM construction.

counting [9]. Other works [4, 23, 46] develop differentially private generative graph models. Specifically, the work [46] focuses on *unattributed* graphs and only capture the structural information of the original graphs. In contrast, the works [4, 23] center on *attributed* graphs and consider capturing the attribute-edge correlation of the original graphs. Chen *et al.* [4] propose CAGM, an extension of [23], which additionally captures the community structure of the original graph. However, as the above methods are designed for the centralized setting, they are not applicable in our decentralized setting. Moreover, the methods focus exclusively on *undirected* graphs, making them unsuitable for achieving satisfactory utility on directed graphs, as shown in our experiments.

Private decentralized graph analytics: Another line of works considers the more challenging *decentralized* setting, where the graph is presented in a decentralized manner, and no party has direct access to the complete graph. Several works [20, 21, 36, 45] focus on extracting differentially private subgraph counts. Other works [3, 33, 44, 48] focus on constructing differentially private generative graph models over decentralized graphs. However, these methods [3, 33, 48] fail to capture the attributed information of the graph. Wei *et al.* [44] propose AsgLDP, which considers decentralized *attributed* graphs by incorporating the attribute-edge correlation into the framework of [33]. AsgLDP is the work most related to ours. However, since it does not consider the directionality of the edges in the input graph, it has poor utility when dealing with directed graphs, as shown in our experiments.

10 CONCLUSION

This paper presents PrivAGM, the first solution for the secure construction of differentially private dAGM on decentralized social graphs. By bridging the gap among edge DP, edge LDP, MPC, and generative graph models, PrivAGM enables aggregators to effectively construct differentially private dAGM on decentralized social graphs without compromising the individual privacy of the users. The evaluation results on three real-world graph datasets demonstrate that PrivAGM outperforms the state-of-the-art baselines, offering significantly higher utility in synthetic graphs.

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A SECURE ATTRIBUTE DISTRIBUTION EXTRACTION

We present how $P_{1,2,3}$ securely extract the noisy version of attribute distribution Θ_F from $\{[\![f_i]\!]\}$, while ensuring that the extracted noisy distribution (denoted as $\widetilde{\Theta}_F$) is differentially private in the individual view of $P_{1,2,3}$. The design intuition is the same as that of Algorithm 4. Specifically, $P_{1,2,3}$ first securely extract the differentially private histogram of attributes, then derive $\widetilde{\Theta}_F$ from the histogram. Algorithm 8 gives our secure attribute extraction protocol. Here, the sensitivity Δ is set to 2, as modifying the attribute of a single vertex leads to changes in the count of two attributes in the social graph.

B PROOF OF THEOREM 1

Proof. Note that the secret shares $\{\langle b_{ij}\rangle\}$ contained within $[E_i]$ reveal nothing about the corresponding values and the adversary does not know the origin of each piece of data. Hence, $[E_i]$ can be described as an observed N-dimensional bit vector

Algorithm 8 Secure Attribute Distribution Extraction

Input: Secret-shared attributes $\{[\![f_i]\!]\}$; privacy budget, parameter ε_F, δ_F.
Output: Differentially private attribute distribution $\widetilde{\Theta}_F$.

// $P_{1,2}$ operate on $\{[\![f_i]\!]\}$:
$[\![\mathbf{T}]\!] = []$. // Empty table.
for all $[\![f_i]\!] \in \{[\![f_i]\!]\}$ **do**
  Append row $[\![f_i]\!] \| [\![\rho]\!]$ to $[\![\mathbf{T}]\!]$, where $[\![\rho]\!] = [\![1]\!]$.
end for
// P_3 constructs a dummy attribute value table $\mathbf{T}'$:
$\mathbf{T}' = []$. // Empty table.
for all $f \in \mathbb{Z}_{2^L}$ **do**
  $n = \max(0, Lap(\varepsilon_F, \delta_F, 2))$. // Draw a noise.
  Append n rows of $f \| \rho$ to $\mathbf{T}'$, where $\rho = 0$.
end for
P_3 secret-shares $\mathbf{T}'$ with $P_{1,2}$ to produce $[\![\mathbf{T}']\!]$.
// P_1, P_2 perform the remaining operations:
$[\![\hat{\mathbf{T}}]\!] = \frac{[\![\mathbf{T}]\!]}{[\![\mathbf{T}']\!]}$. // Vertical concatenation.
$[\![\hat{\mathbf{T}}']\!] = \mathsf{secShuffle}([\![\hat{\mathbf{T}}]\!])$. //Oblivious row-wise shuffle.
Safely reveal column 1 of $[\![\hat{\mathbf{T}}']\!]$ to obtain $[\hat{\mathbf{T}}']$.
Initialize secret-shared histogram $[\![\mathbf{H}]\!] = [\![0]\!]^{2^L}$.
for all row $f \| [\![\rho]\!]$ in $[\hat{\mathbf{T}}']$ **do**
  $[\![\mathbf{H}[f]]\!] = [\![\mathbf{H}[f]]\!] + [\![\rho]\!]$. // Index attribute value to histogram.
end for
$[\![\widetilde{\mathbf{H}}[f]]\!] += [\![Lap(\varepsilon_F, 0, 2)]\!], f \in \mathbb{Z}_{2^L}$.
Safely reveal the differentially private histogram $\widetilde{\mathbf{H}}$.
$\widetilde{\mathbf{H}}'[f] = \max(0, \widetilde{\mathbf{H}}[f]), f \in \mathbb{Z}_{2^L}$. // Truncation.
$\widetilde{\mathbf{H}}''[f] = \frac{\widetilde{\mathbf{H}}'[f]}{\mathsf{sum}(\widetilde{\mathbf{H}}')}, f \in \mathbb{Z}_{2^L}$. // Normalization
Initialize the attribute distribution $\widetilde{\Theta}_F = \{0\}^{2^L}$.
$\widetilde{\Theta}_F(f) = \widetilde{\mathbf{H}}''[f], f \in \mathbb{Z}_{2^L}$.
return Differentially private attribute distribution $\widetilde{\Theta}_F$.

$\mathbf{b} = (b_1, \cdots, b_N)$ for the adversary, where $b_j = 1, j \in [1, N]$ if $(i, j, \langle b_{ij} \rangle) \in [E_i]$, and otherwise $b_j = 0$.

We abuse $\mathcal{M}$ to denote the secure mechanisms in PrivAGM, and use $Pr[\mathbf{A}[i, j] \to b_j]$ to represent the probability that, given a bit $\mathbf{A}[i, j]$ of user i's local neighbor list, the adversary observes $b_j \in \{0, 1\}$. The values of $Pr[\mathbf{A}[i, j] \to b_j]$ can be categorized into four cases:

• $\mathbf{A}[i, j] = 0$ and $b_j = 0$: This case corresponds to the scenario where either user i does not include the zero bit $\mathbf{A}[i, j]$ in E_i (with a probability of $1-p$), or user i includes $\mathbf{A}[i, j]$ in E_i (with a probability of p), but does not choose the adversary to receive $(i, j, \langle 0 \rangle)$ (with a probability of $1/3$). Therefore, $Pr[0 \to 0] = 1-p+p \cdot 1/3 = 1-2p/3$.
• $\mathbf{A}[i, j] = 1$ and $b_j = 0$: This case corresponds to the scenario where user i includes the nonzero bit $\mathbf{A}[i, j]$ in E_i (with a probability of 1), but does not choose the adversary to receive $(i, j, \langle 1 \rangle)$ (with a probability of $1/3$). Therefore, $Pr[1 \to 0] = 1 \cdot 1/3 = 1/3$.
• $\mathbf{A}[i, j] = 1$ and $b_j = 1$: This case corresponds to the scenario where user i includes the nonzero bit $\mathbf{A}[i, j]$ in E_i (with a probability of 1) and also chooses the adversary to receive $(i, j, \langle 1 \rangle)$ (with a probability of $2/3$). Therefore, $Pr[1 \to 1] = 1 \cdot 2/3 = 2/3$.
• $\mathbf{A}[i, j] = 0$ and $b_j = 1$: This case corresponds to the scenario where user i includes the zero bit $\mathbf{A}[i, j]$ in E_i (with a probability of p) and also chooses the adversary to receive $(i, j, \langle 0 \rangle)$ (with a probability of $2/3$). Therefore, $Pr[0 \to 1] = p \cdot 2/3 = 2p/3$.

Given any two local neighbor lists $\mathbf{A}[i, :]$ and $\mathbf{A}'[i, :]$ that differ in one bit, we need to prove that $\frac{Pr[\mathcal{M}(\mathbf{A}[i,:])=\mathbf{b}]}{Pr[\mathcal{M}(\mathbf{A}'[i,:])=\mathbf{b}]} \leq e^{\varepsilon_E}$, where $\varepsilon_E = \max\{\ln(1/p), \ln(3-2p)\}$. Without loss of generality, let us assume that $\mathbf{A}[i, 1]$ and $\mathbf{A}'[i, 1]$ are not identical. Then we have:

$$\frac{Pr[\mathcal{M}(\mathbf{A}[i,:])=\mathbf{b}]}{Pr[\mathcal{M}(\mathbf{A}'[i,:])=\mathbf{b}]} = \frac{Pr[\mathbf{A}[i,1] \to b_1] \cdots Pr[\mathbf{A}[i,N] \to b_N]}{Pr[\mathbf{A}'[i,1] \to b_1] \cdots Pr[\mathbf{A}'[i,N] \to b_N]} = \frac{Pr[\mathbf{A}[i,1] \to b_1]}{Pr[\mathbf{A}'[i,1] \to b_1]}. \tag{3}$$

We analyze Eq. 3 in four cases:
1) $\frac{Pr[0\to0]}{Pr[1\to0]} = \frac{1-2p/3}{1/3} = 3-2p$; 2) $\frac{Pr[1\to0]}{Pr[0\to0]} = \frac{1/3}{1-2p/3} = \frac{1}{3-2p}$;
3) $\frac{Pr[0\to1]}{Pr[1\to1]} = \frac{2p/3}{2/3} = p$; 4) $\frac{Pr[1\to1]}{Pr[0\to1]} = \frac{2/3}{2p/3} = 1/p$.
We then need to find the upper bound of the set $\{3-2p, \frac{1}{3-2p}, p, 1/p\}$. Since $0 \leq p \leq 1$, we have $1/p \geq p$ and $3-2p \geq \frac{1}{3-2p}$. Therefore, we have $\{3-2p, \frac{1}{3-2p}, p, 1/p\} \leq \max\{1/p, 3-2p\} = e^{\varepsilon_E}$, and $\varepsilon_E = \max\{\ln(1/p), \ln(3-2p)\}$. The probability that the adversary views the same local neighbor list $\mathbf{b}$ from $\mathbf{A}[i, :], \mathbf{A}'[i, :]$ is bounded by ε_E, which satisfies ε_E-edge LDP based on Definition 1. □

C REDUCING ε_E WITH MORE AGGREGATORS

We provide a proof demonstrating that we can further diminish ε_E given in Theorem 1 by introducing more aggregators.

Theorem 7. *The local neighbor list $[E_i] \in \mathcal{L}_E$ of user $i \in [1, N]$ held by the adversary adheres to pure ε_E-edge LDP with $\varepsilon_E = \max\{\ln(1/p), \ln(V-(V-2) \cdot p)\}$, where p denotes the selective probability of each 0 element in user i's local neighbor list and V ($V \geq 3$) denotes the number of aggregators.*

Proof. Note that the secret shares $\{\langle b_{ij} \rangle\}$ contained within $[E_i]$ reveal nothing about the corresponding values and the adversary does not know the origin of each piece of data. Hence, $[E_i]$ can be described as an observed N-dimensional bit vector $\mathbf{b} = (b_1, \cdots, b_N)$ for the adversary, where $b_j = 1, j \in [1, N]$ if $(i, j, \langle b_{ij} \rangle) \in [E_i]$, and otherwise $b_j = 0$.

We abuse $\mathcal{M}$ to denote the secure mechanisms in PrivAGM, and use $Pr[\mathbf{A}[i, j] \to b_j]$ to represent the probability that, given a bit $\mathbf{A}[i, j]$ of user i's local neighbor list, the adversary observes $b_j \in \{0, 1\}$. The values of $Pr[\mathbf{A}[i, j] \to b_j]$ can be categorized into four cases:

• $\mathbf{A}[i, j] = 0$ and $b_j = 0$: This case corresponds to the scenario where either user i does not include the zero bit $\mathbf{A}[i, j]$ in E_i (with a probability of $1-p$), or user i includes $\mathbf{A}[i, j]$ in E_i (with a probability of p), but does not choose the adversary to receive $(i, j, \langle 0 \rangle)$ (with a probability of $\frac{V-2}{V}$). Therefore, $Pr[0 \to 0] = 1-p+p \cdot \frac{V-2}{V} = 1-\frac{2p}{V}$.
• $\mathbf{A}[i, j] = 1$ and $b_j = 0$: This case corresponds to the scenario where user i includes the nonzero bit $\mathbf{A}[i, j]$ in E_i (with a probability of 1), but does not choose the adversary to receive $(i, j, \langle 1 \rangle)$ (with a probability of $\frac{V-2}{V}$). Therefore, $Pr[1 \to 0] = 1 \cdot \frac{V-2}{V} = \frac{V-2}{V}$.
• $\mathbf{A}[i, j] = 1$ and $b_j = 1$: This case corresponds to the scenario where user i includes the nonzero bit $\mathbf{A}[i, j]$ in E_i (with a probability

of 1) and also chooses the adversary to receive $(i, j, \langle 1 \rangle)$ (with a probability of $\frac{2}{V}$). Therefore, $Pr[1 \to 1] = 1 \cdot \frac{2}{V} = \frac{2}{V}$.

• $\mathbf{A}[i, j] = 0$ and $b_j = 1$: This case corresponds to the scenario where user i includes the zero bit $\mathbf{A}[i, j]$ in E_i (with a probability of p) and also chooses the adversary to receive $(i, j, \langle 0 \rangle)$ (with a probability of $\frac{2}{V}$). Therefore, $Pr[0 \to 1] = p \cdot \frac{2}{V} = \frac{2p}{V}$.

Given any two local neighbor lists $\mathbf{A}[i, :]$ and $\mathbf{A}'[i, :]$ that differ in one bit, we need to prove that $\frac{Pr[\mathcal{M}(\mathbf{A}[i,:])=\mathbf{b}]}{Pr[\mathcal{M}(\mathbf{A}'[i,:])=\mathbf{b}]} \leq e^{\varepsilon_E}$, where $\varepsilon_E = \max\{\ln(1/p), \ln((V-2p)/(V-2))\}$. Without loss of generality, let us assume that $\mathbf{A}[i, 1]$ and $\mathbf{A}'[i, 1]$ are not identical. Then we have:

$$\frac{Pr[\mathcal{M}(\mathbf{A}[i,:])=\mathbf{b}]}{Pr[\mathcal{M}(\mathbf{A}'[i,:])=\mathbf{b}]} = \frac{Pr[\mathbf{A}[i,1] \to b_1] \cdots Pr[\mathbf{A}[i,N] \to b_N]}{Pr[\mathbf{A}'[i,1] \to b_1] \cdots Pr[\mathbf{A}'[i,N] \to b_N]} = \frac{Pr[\mathbf{A}[i,1] \to b_1]}{Pr[\mathbf{A}'[i,1] \to b_1]}. \tag{4}$$

We analyze Eq. 4 in four cases:
1) $\frac{Pr[0\to0]}{Pr[1\to0]} = \frac{(V-2p)/V}{(V-2)/V} = \frac{V-2p}{V-2}$; 2) $\frac{Pr[1\to0]}{Pr[0\to0]} = \frac{(V-2)/V}{(V-2p)/V} = \frac{V-2}{V-2p}$; 3) $\frac{Pr[0\to1]}{Pr[1\to1]} = \frac{2p/V}{2/V} = p$; 4) $\frac{Pr[1\to1]}{Pr[0\to1]} = \frac{2/V}{2p/V} = 1/p$.
We then need to find the upper bound of $\{\frac{V-2p}{V-2}, \frac{V-2}{V-2p}, p, 1/p\}$. Since $0 \leq p \leq 1$, we have $1/p \geq p$ and $\frac{V-2p}{V-2} \geq \frac{V-2}{V-2p}$. Therefore, we can conclude that $\{\frac{V-2p}{V-2}, \frac{V-2}{V-2p}, p, 1/p\} \leq \max\{1/p, \frac{V-2p}{V-2}\} = e^{\varepsilon_E}$, and $\varepsilon_E = \max\{\ln(1/p), \ln((V-2p)/(V-2))\}$. The probability that the adversary views the local neighbor list $\mathbf{b}$ from $\mathbf{A}[i, :]$ and $\mathbf{A}'[i, :]$ is bounded by ε_E, which satisfies ε_E-edge LDP based on Def. 1. □

Given that $0 \leq p \leq 1$, we have $\max\{\ln(1/p), \ln((V-2p)/(V-2))\} \leq \max\{\ln(1/p), \ln(3-2p)\}$. This demonstrates that we can reduce ε_E by introducing additional aggregators.

D PROOF OF THEOREM 2

Proof. Note that only $P_{1,2}$ have access to $[\hat{\mathbf{T}}'_{\mathcal{D}}]$, and therefore we only analyze the view of the adversary who controls either P_1 or P_2. Since the secret shares (i.e., the 3nd column of table $[\hat{\mathbf{T}}'_{\mathcal{D}}]$) reveal nothing about the corresponding values, $[\hat{\mathbf{T}}'_{\mathcal{D}}]$ can be described as a histogram (denoted as $\mathbf{H}_{\mathcal{D}}$) of degree pairs observed by the adversary. In $\mathbf{H}_{\mathcal{D}}$, each bin $\mathbf{H}_{\mathcal{D}}[d^+, d^-], d^+, d^- \in [0, k]$ represents the (noisy) number of vertices that have degree pair (d^+, d^-). Then we proceed to prove that with probability $1-\delta_{\mathcal{D}}$, the probability for the adversary viewing the same $\mathbf{H}_{\mathcal{D}}$ from two neighboring graphs G and G' is bounded by $e^{\varepsilon_{\mathcal{D}}}$.

If all the noise (drawn in line 8 of Algorithm 3) for the degree pairs are non-negative, then the probability to output the same noisy histogram $\mathbf{H}_{\mathcal{D}}$ from G, G' is $\frac{Pr[\mathcal{M}(G)=\mathbf{H}_{\mathcal{D}}]}{Pr[\mathcal{M}(G')=\mathbf{H}_{\mathcal{D}}]}$, which equals

$$\frac{\prod_{d^+,d^- \in [0,k]} Pr[\mathbf{H}_{\mathcal{D}}[d^+, d^-] - \mathbf{H}_G[d^+, d^-]]}{\prod_{d^+,d^- \in [0,k]} Pr[\mathbf{H}_{\mathcal{D}}[d^+, d^-] - \mathbf{H}_{G'}[d^+, d^-]]}, \tag{5}$$

where $\mathbf{H}_G[d^+, d^-]$ is the number of vertices in graph G that have degree pair (d^+, d^-). $Pr[\mathbf{H}_{\mathcal{D}}[d^+, d^-] - \mathbf{H}_G[d^+, d^-]]$ is the probability drawing the noise $\mathbf{H}_{\mathcal{D}}[d^+, d^-] - \mathbf{H}_G[d^+, d^-]$ from the Laplace distribution $Lap(\varepsilon_{\mathcal{D}}, \delta_{\mathcal{D}}, 4)$. Let D_Δ be the set of bins where $\mathbf{H}_G$ and $\mathbf{H}_G$ are not equal. Then we can convert Eq. 5 as follows:

$$\begin{aligned}
&\frac{\prod_{(d^+,d^-)\in D_\Delta} e^{\frac{-\varepsilon_{\mathcal{D}}}{4} \cdot |\mathbf{H}_{\mathcal{D}}[d^+,d^-] - \mathbf{H}_G[d^+,d^-] - \mu|}}{\prod_{(d^+,d^-)\in D_\Delta} e^{\frac{-\varepsilon_{\mathcal{D}}}{4} \cdot |\mathbf{H}_{\mathcal{D}}[d^+,d^-] - \mathbf{H}_{G'}[d^+,d^-] - \mu|}} \\
&= \prod_{(d^+,d^-)\in D_\Delta} e^{\frac{\varepsilon_{\mathcal{D}}}{4} \cdot (|\mathbf{H}_{\mathcal{D}}[d^+,d^-] - \mathbf{H}_{G'}[d^+,d^-] - \mu| - |\mathbf{H}_{\mathcal{D}}[d^+,d^-] - \mathbf{H}_G[d^+,d^-] - \mu|)} \\
&= e^{\frac{\varepsilon_{\mathcal{D}}}{4} \cdot \sum_{(d^+,d^-)\in D_\Delta} |\mathbf{H}_{\mathcal{D}}[d^+,d^-] - \mathbf{H}_{G'}[d^+,d^-] - \mu| - |\mathbf{H}_{\mathcal{D}}[d^+,d^-] - \mathbf{H}_G[d^+,d^-] - \mu|} \\
&\leq e^{\frac{\varepsilon_{\mathcal{D}}}{4} \cdot \sum_{(d^+,d^-)\in D_\Delta} |\mathbf{H}_G[d^+,d^-] - \mathbf{H}_{G'}[d^+,d^-]|} \leq e^{\frac{\varepsilon_{\mathcal{D}}}{4} \cdot 4} = e^{\varepsilon_{\mathcal{D}}}.
\end{aligned} \tag{6}$$

The above proof is based on a fact that as G and G' differ by the presence or absence of a single edge, the maximum discrepancy in the values of their degree pair histograms is 4, i.e., $\sum_{(d^+,d^-)\in D_\Delta} |\mathbf{H}_G[d^+, d^-] - \mathbf{H}_{G'}[d^+, d^-]| \leq 4$. This is because the addition or removal of an edge alters the out-degree and in-degree of exactly two vertices, which can potentially result in changes of up to four degree pairs with a frequency difference of 1.

The probability to draw a negative noise from $Lap(\varepsilon_{\mathcal{D}}, \delta_{\mathcal{D}}, 4)$ is [18]: $Pr[n < 0] = \sum_{x=-1}^{-\infty} \frac{e^{\frac{\varepsilon_{\mathcal{D}}}{4}} - 1}{e^{\frac{\varepsilon_{\mathcal{D}}}{4}} + 1} \cdot e^{\frac{-\varepsilon_{\mathcal{D}} \cdot |x-\mu|}{4}} = \frac{e^{\frac{-\mu \cdot \varepsilon_{\mathcal{D}}}{4}}}{e^{\frac{\varepsilon_{\mathcal{D}}}{4}} + 1}$. Given $\mu = -\frac{4}{\varepsilon_{\mathcal{D}}} \cdot \ln((e^{\frac{\varepsilon_{\mathcal{D}}}{4}} + 1) \cdot (1 - (1-\delta_{\mathcal{D}})^{\frac{1}{4}}))$, we have $Pr[n < 0] = 1 - (1-\delta_{\mathcal{D}})^{\frac{1}{4}}$. As there are at most 4 bins in $\mathbf{H}_G, \mathbf{H}_{G'}$ that are not equal. Hence, the overall failing probability is $1 - (1 - Pr[n < 0])^4 = \delta_{\mathcal{D}}$. This means that with probability $1 - \delta_{\mathcal{D}}$, the probability for the adversary viewing the same $\mathbf{H}_{\mathcal{D}}$ from G, G' is bounded by $e^{\varepsilon_{\mathcal{D}}}$, which satisfies $(\varepsilon_{\mathcal{D}}, \delta_{\mathcal{D}})$-edge DP based on Definition 2.

Then we proceed to prove that the probability for the adversary viewing the same $\widetilde{\mathbf{H}}_{\mathcal{D}}$ from two neighboring graphs G and G' is bounded by $e^{\varepsilon_{\mathcal{D}}}$. The probability to output the same $\widetilde{\mathbf{H}}_{\mathcal{D}}$ from G and G' is bounded by $\frac{Pr[\mathcal{M}(G)=\widetilde{\mathbf{H}}_{\mathcal{D}}]}{Pr[\mathcal{M}(G')=\widetilde{\mathbf{H}}_{\mathcal{D}}]} \leq e^{\varepsilon_{\mathcal{D}}}$. The proof follows Eq. 6 directly. Since the noises added to $\widetilde{\mathbf{H}}_{\mathcal{D}}$ (line 19 of Algorithm 3) are drawn from $Lap(\varepsilon_{\mathcal{D}}, 0, 4)$ (i.e., $\delta = 0$), $\mathcal{M}$ satisfies pure $\varepsilon_{\mathcal{D}}$-edge DP based on Definition 2. □

E PROOF OF THEOREM 5

Proof. First, we establish the ε_Δ-edge DP of $\widetilde{n}^A_\Delta$, and the same conclusion holds for $\widetilde{n}^B_\Delta$. Given two neighboring graphs G, G', we have $|n^A_\Delta - n'^A_\Delta| \leq k-1$, where n^A_Δ and n'^A_Δ represent the count of triangle TriA in G, G' respectively, after edge clipping with parameter k. The probability to output the same noisy triangle TriA count $\widetilde{n}^A_\Delta$ from G, G' is bounded by

$$\begin{aligned}
&\frac{Pr[\mathcal{M}(G) = \widetilde{n}^A_\Delta]}{Pr[\mathcal{M}(G') = \widetilde{n}^A_\Delta]} = \frac{Pr[n^A_\Delta - \widetilde{n}^A_\Delta]}{Pr[n'^A_\Delta - \widetilde{n}^A_\Delta]} \\
&= \frac{e^{\frac{-\varepsilon_\Delta \cdot |n^A_\Delta - \widetilde{n}^A_\Delta|}{k-1}}}{e^{\frac{-\varepsilon_\Delta \cdot |n'^A_\Delta - \widetilde{n}^A_\Delta|}{k-1}}} = e^{\frac{\varepsilon_\Delta}{k-1} \cdot (|n'^A_\Delta - \widetilde{n}^A_\Delta| - |n^A_\Delta - \widetilde{n}^A_\Delta|)} \\
&\leq e^{\frac{\varepsilon_\Delta}{k-1} \cdot (|n'^A_\Delta - n^A_\Delta|)} \leq e^{\varepsilon_\Delta}.
\end{aligned}$$

$\mathcal{M}$ satisfies ε_Δ-edge DP for $\widetilde{n}^A_\Delta$ based on Definition 2. □

F PROOF OF THEOREM 6

Proof. As no user receives any values from other parties, the simulator for the users must exist. Therefore, we only analyze the existence of the simulator for $P_{\{1,2,3\}}$. Since the roles of $P_{\{1,2,3\}}$ in PrivAGM are symmetric, with the exception that $P_{1,2}$ receive the secret shares of dummy entities (i.e., $[\![\mathbf{T}']\!]$) from P_3, it is enough to analyze the existence of $\mathcal{S}$ under the condition that $\mathcal{A}$ corrupts P_1.

Simulator $\mathcal{S}$ in Colle**.** During this phase, P_1 exclusively receives the set of tuples $\{(i, j, \langle b_{ij}\rangle)\}$. Therefore, we can trivially construct the simulator $\mathcal{S}$ by invoking the simulator of ASS [7] and providing it with the output of $\mathcal{L}_E$ as input.

Simulator $\mathcal{S}$ in edgeClip**.** The efficient oblivious edge clipping protocol is presented in Algorithm 2. Initially, $\mathcal{A}$ possesses the clipped local neighbor lists $\{(i, j, \langle b_{ij}\rangle)\}$, k, and w. $\mathcal{A}$ only receives messages from other honest aggregators during the execution of lines 7 and 10. The operations in line 7 is based on the FSS-based DCF from [2]. Hence, the simulator can be trivially constructed by invoking the simulator of FSS-based DCF [2] and providing it with $k/3$ as input. The operation in line 10 relies on the basic secure multiplication operation in the ASS domain. Hence, the simulator can be trivially constructed by invoking the simulator of ASS [7].

Simulator $\mathcal{S}$ in extDeg**.** The secure degree sequence extraction protocol is given in Algorithm 3. Initially, $\mathcal{A}$ possesses the clipped local neighbor lists $\{(i, j, \langle b_{ij}\rangle)\}$, k, $\varepsilon_{\mathcal{D}}$, $\delta_{\mathcal{D}}$, and the secret shares of Laplacian noises. $\mathcal{A}$ only receives messages from other honest aggregators during the execution of lines 3, 11, 13, 14, and 20. The operations in lines 3 and 11 rely on the secret sharing operation in the ASS domain. Hence, the simulator can be trivially constructed by invoking the simulator of ASS [7]. The operation in line 13 is based on the oblivious shuffle protocol from [14]. Therefore, the simulator can be trivially constructed by invoking the simulator of oblivious shuffle [14]. During the execution of lines 14 and 20, $\mathcal{A}$ receives secret shares from P_2 to reveal $[\hat{\mathbf{T}}']$ and $\widetilde{\mathbf{H}}$. Based on the security of ASS, the secret shares from P_2 are uniformly random in the view of $\mathcal{A}$. For revealed $[\hat{\mathbf{T}}']$ and $\widetilde{\mathbf{H}}$, $\mathcal{S}$ adjusts P_2' shares such that the revealed values are indeed the coefficient vector it received from $\mathcal{F}$ (i.e., $\mathcal{L}_{\mathcal{D}}$).

Simulator $\mathcal{S}$ in extCorr **and** extAttr**.** As the analysis of the existence of $\mathcal{S}$ in extCorr and extAttr follows the same idea as that in extDeg, we omit the analysis here.

Simulator $\mathcal{S}$ in extTria**.** The secure triangle counts extraction protocol is given in Algorithm 5. Initially, $\mathcal{A}$ possesses $\{(i, j, \langle b_{ij}\rangle)\}$, k, $\varepsilon_{\mathcal{D}}$, $\delta_{\mathcal{D}}$, and the secret shares of Laplacian noises. $\mathcal{A}$ only receives messages from $P_{2,3}$ during the execution of lines 6, 14, and 22. The operations in lines 6 and 14 are based on the secure 3-input multiplication gate from [31]. Hence, the simulator can be trivially constructed by invoking the simulator of the secure 3-input multiplication gate [31]. During the execution of line 22, $\mathcal{A}$ receives secret shares from P_2 and P_3 to reveal $\widetilde{n}^A_{\triangle}$ and $\widetilde{n}^B_{\triangle}$. Based on the security of ASS, the secret shares from $P_{2,3}$ are uniformly random in the view of $\mathcal{A}$. For revealed $\widetilde{n}^A_{\triangle}, \widetilde{n}^B_{\triangle}$, $\mathcal{S}$ adjusts the shares of $P_{2,3}$ such that the revealed values are indeed the coefficient vector it received from $\mathcal{F}$ (i.e., $\mathcal{L}_{\triangle}$). □

G UTILITY METRICS

We evaluate the utility of the synthetic graphs using four widely-used metrics [4, 23, 33, 44].

• *Degree distribution:* We utilize the Kolmogorov-Smirnov statistic to evaluate how well a synthetic graph captures the degree distribution of the original graph (denoted as $\mathsf{KS}_{\mathsf{degree}}$), which quantifies the maximum distance between two distributions. Let $c_{\mathcal{D}^{\pm}}$ and $c'_{\mathcal{D}^{\pm}}$ represent the cumulative distribution functions estimated from the sorted degree pair sequences of the input and synthetic graphs, respectively. Then $\mathsf{KS}_{\mathsf{degree}} = \max_{d^+=0}^{d^+_{max}} \max_{d^-=0}^{d^-_{max}} |c_{\mathcal{D}^{\pm}}(d^+, d^-) - c'_{\mathcal{D}^{\pm}}(d^+, d^-)|$, where d^+_{max}/d^-_{max} are the maximum out/in-degree in the input and synthetic graphs. $c_{\mathcal{D}^{\pm}}(d^+, d^-)$ (and $c'_{\mathcal{D}^{\pm}}(d^+, d^-)$) represents the percentage of vertices in the input (and synthetic) graph whose out-degrees $\leq d^+$ and in-degrees $\leq d^-$. As $\mathsf{KS}_{\mathsf{degree}}$ is less sensitive to differences in the tails of the degree distributions, we additionally report the Hellinger distance between two degree distributions $\Theta_{\mathcal{D}^{\pm}}$ and $\Theta'_{\mathcal{D}^{\pm}}$:

$$\mathsf{HD}_{\mathsf{degree}} = \frac{1}{\sqrt{2}} \cdot \sqrt{\sum_{d^+=0}^{d^+_{max}} \sum_{d^-=0}^{d^-_{max}} (\sqrt{\Theta_{\mathcal{D}^{\pm}}(d^+, d^-)} - \sqrt{\Theta'_{\mathcal{D}^{\pm}}(d^+, d^-)})^2},$$

where $\Theta_{\mathcal{D}^{\pm}}(d^+, d^-)$ is the percentage of vertices having a degree pair (d^+, d^-) in the graph. A smaller $\mathsf{KS}_{\mathsf{degree}}$ and $\mathsf{HD}_{\mathsf{degree}}$ indicate a closer alignment between the degree distributions of the synthetic and original graphs.

• *Attribute-edge correlation distribution:* To evaluate how well a synthetic graph captures the attribute-edge correlations of the original graph, we report the mean relative error (MRE) between their attribute-edge correlation distributions (denoted as $\mathsf{MRE}_{\mathsf{corr}}$), and the Hellinger distance:

$$\mathsf{HD}_{\mathsf{corr}} = \frac{1}{\sqrt{2}} \cdot \sqrt{\sum_{f=0}^{2^L} \sum_{f'=0}^{2^L} (\sqrt{\Theta_{\vec{X}}(f \to f')} - \sqrt{\Theta'_{\vec{X}}(f \to f')})^2},$$

where $\Theta_{\vec{X}}(f \to f')$ is the proportion of edges that connect a vertex with attribute value f to a vertex with attribute value f' in a graph.

• *Local clustering coefficient:* The local clustering coefficient of a vertex in a graph quantifies how close its neighbors are to forming a clique. The local clustering coefficient C_i for a vertex i is determined by the proportion of edges between the vertices within its neighborhood divided by the number of edges that could potentially exist between them. In directed graphs, $C_i = \frac{|\{(j,k):j,k \in Ne_i, (j,k) \in E\}|}{(d_i^+ + d_i^-) \cdot (d_i^+ + d_i^- - 1)}$, where Ne_i is the set of vertex i's out/in-neighbors, E is the edge set, and (j, k) represents the edge from vertex j to vertex k [43]. The average of the local clustering coefficient is $C = \frac{\sum_{i \in [1,N]} C_i}{N}$. We report the MRE between the average of the local clustering coefficient of the synthetic and original graphs (denoted as $\mathsf{MRE}_{\mathsf{cluster}}$).

• *Triangle counts:* We report the MRE in the number of triangles, TriA and TriB, between the synthetic and original graphs (denoted as $\mathsf{MRE}_{\mathsf{TriA}}$ and $\mathsf{MRE}_{\mathsf{TriB}}$ respectively).